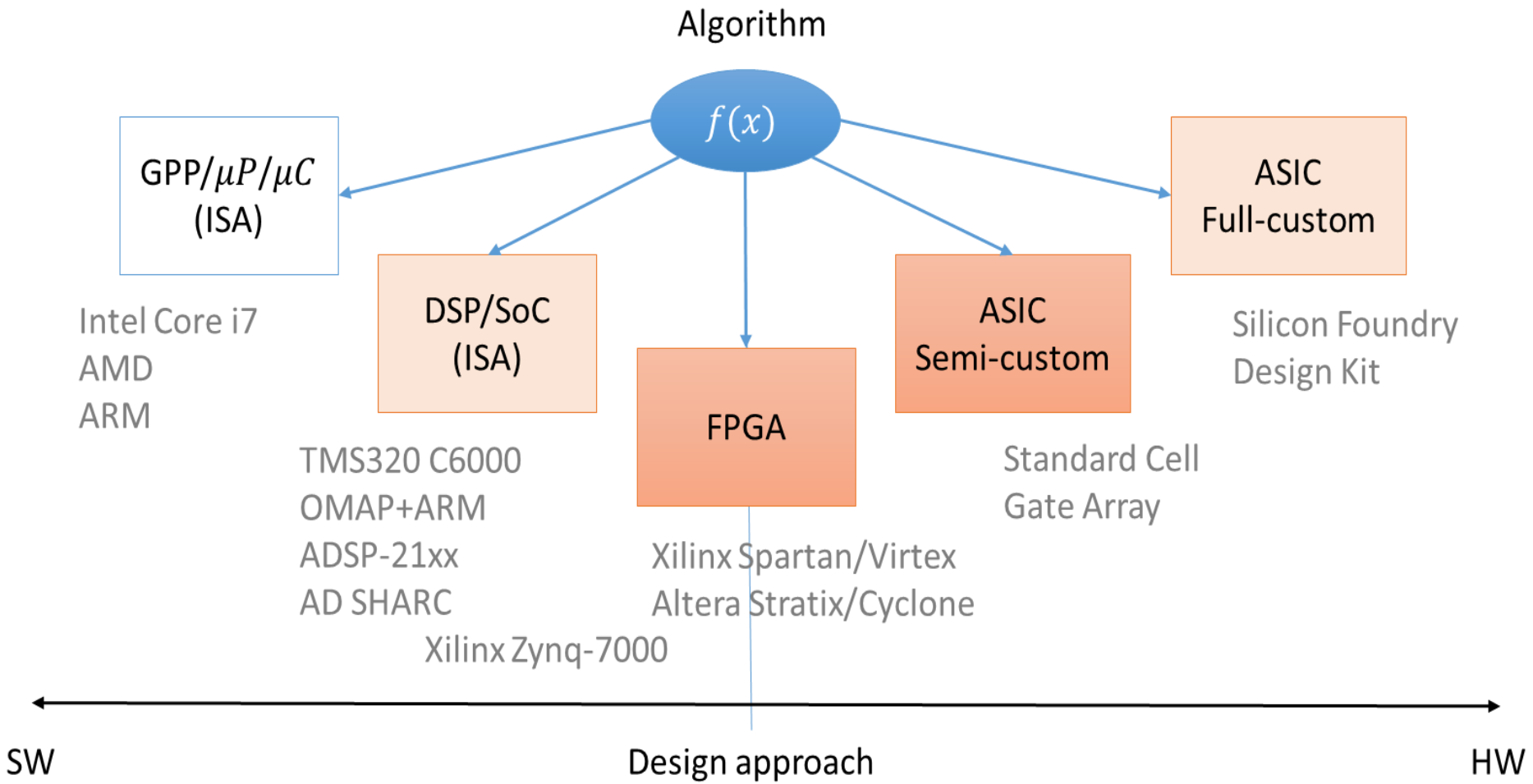
Electronics and Communication Systems
Electronics Systems

Master Degree in Computer Engineering

https://computer.ing.unipi.it/ce-lm

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Email: luca.fanucci@unipi.it

How to map a system/algorithm to hw/sw design



Outline

- Review
  - Sync. and Async Sequential Logic
  - Finite State Machine (MEALY, MOORE, MEALY Rit.)
- Complementary CMOS Logic
  - Synthesis
  - Dynamic Analysis
  - Examples: NAND, NOR and comparison
  - Stick Diagram
  - Body Effect
  - Standard-cell AMS 0,35 um
- Example: Inverter Chain
  - Driving an high value capacitor
  - Dynamic Analysis
  - Power-Speed Trade-Off
- Example: Pass Transistor Chain
  - Dynamic Analysis
  - Area – Speed Trade-Off

Logic

- Combinational Logic
  - Output are a pure function of the present input only
- Sequential Logic
  - Output depends not only on the present input but also on the history of the input

Sequential Logic

- MEMORY concept
- Asynchronous sequential
  - Outputs of the circuit change directly in response to changes in inputs
- Synchronous sequential
  - Outputs of the circuit change only with the rising (falling) edge of a reference control signal, named clock

Asynchronous sequential

- Ripple Design
- Pro
  - High Speed
  - Low Power
- Cont
  - Race Conditions
  - Difficult Design

Synchronous sequential

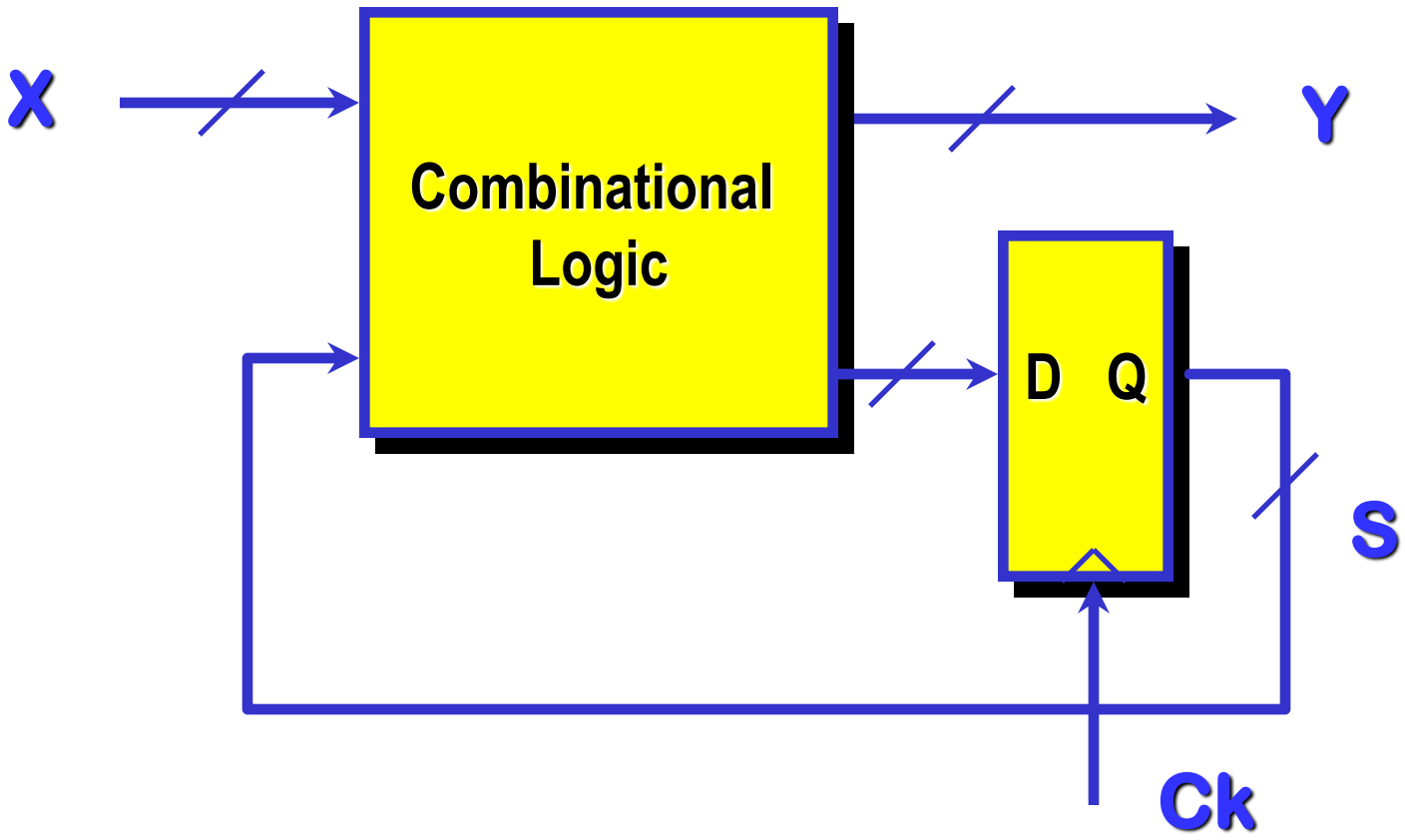
- Clocked Design
- Pro
  - Easy Design
  - Robust Design
- Cont
  - High complexity
  - High Power Consumption
  - « Reduced » Speed

MEALY Machine1/2

- Output values are determined both by its current state and the current inputs
- Asynchronous Logic
- Based on one combinational Logic

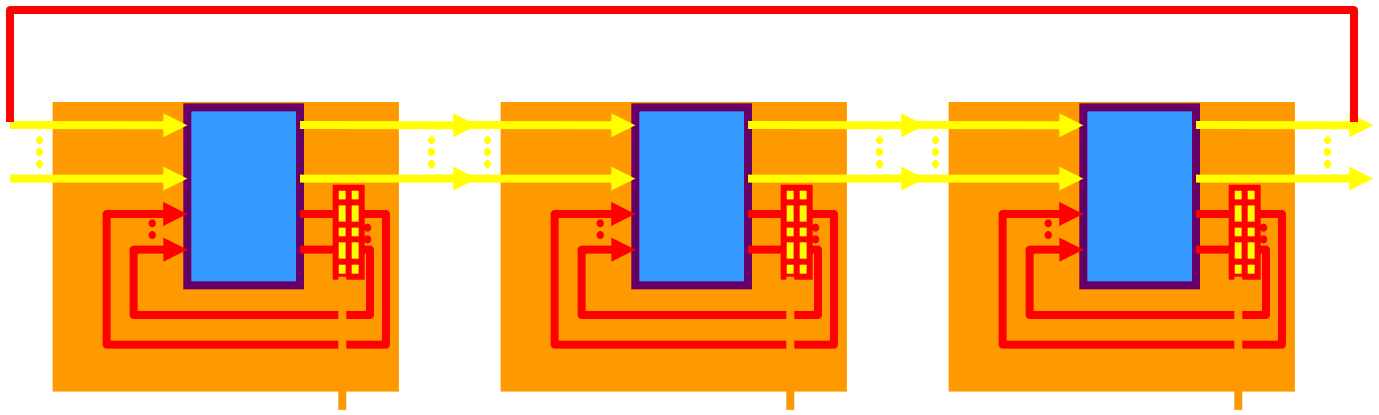
MEALY Machine 2/2

- Block Diagram



Notes

- Outputs are asynchronous
- Critical to have a cascaded of MEALY machines due to hidden asynchronous logic

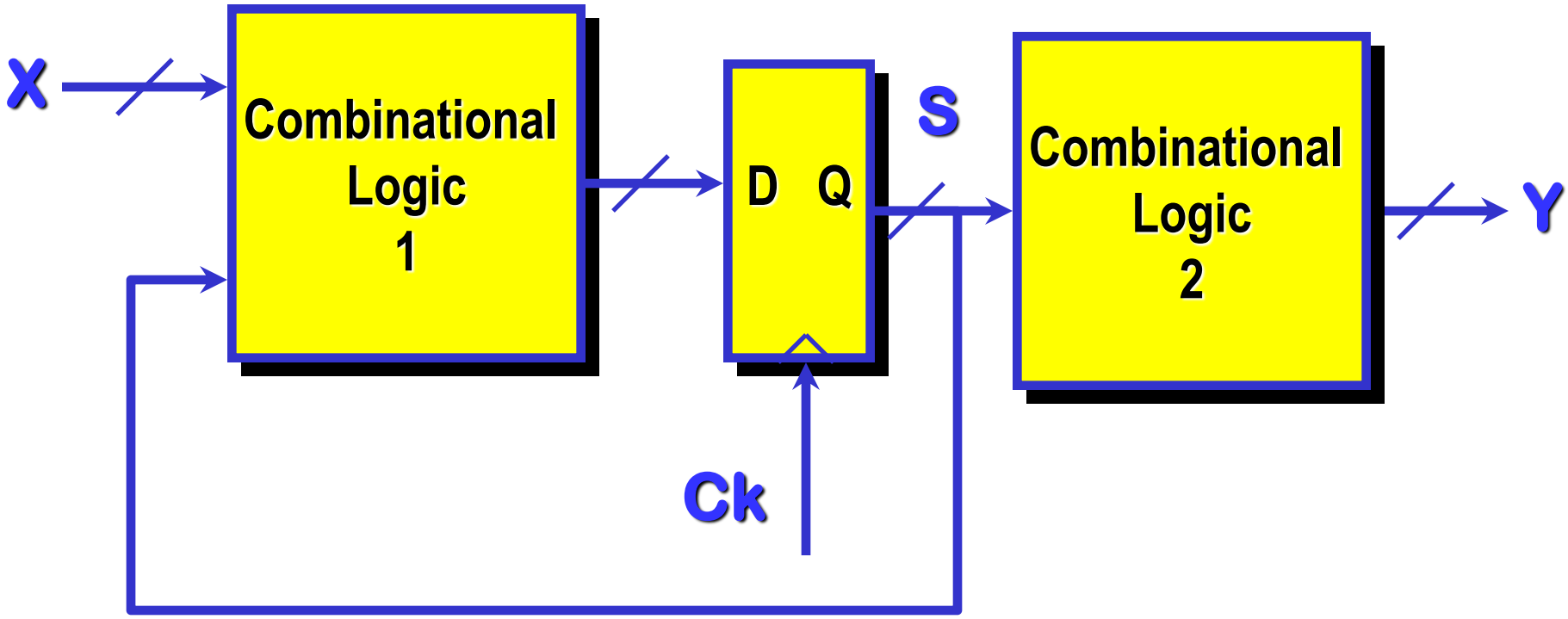


MOORE Machine 1/2

- output values are determined solely by its current state
- Synchrounous Logic
- Required two Combinational Logics
- Never Unstable

MOORE Machine 2/2

✧Block diagram

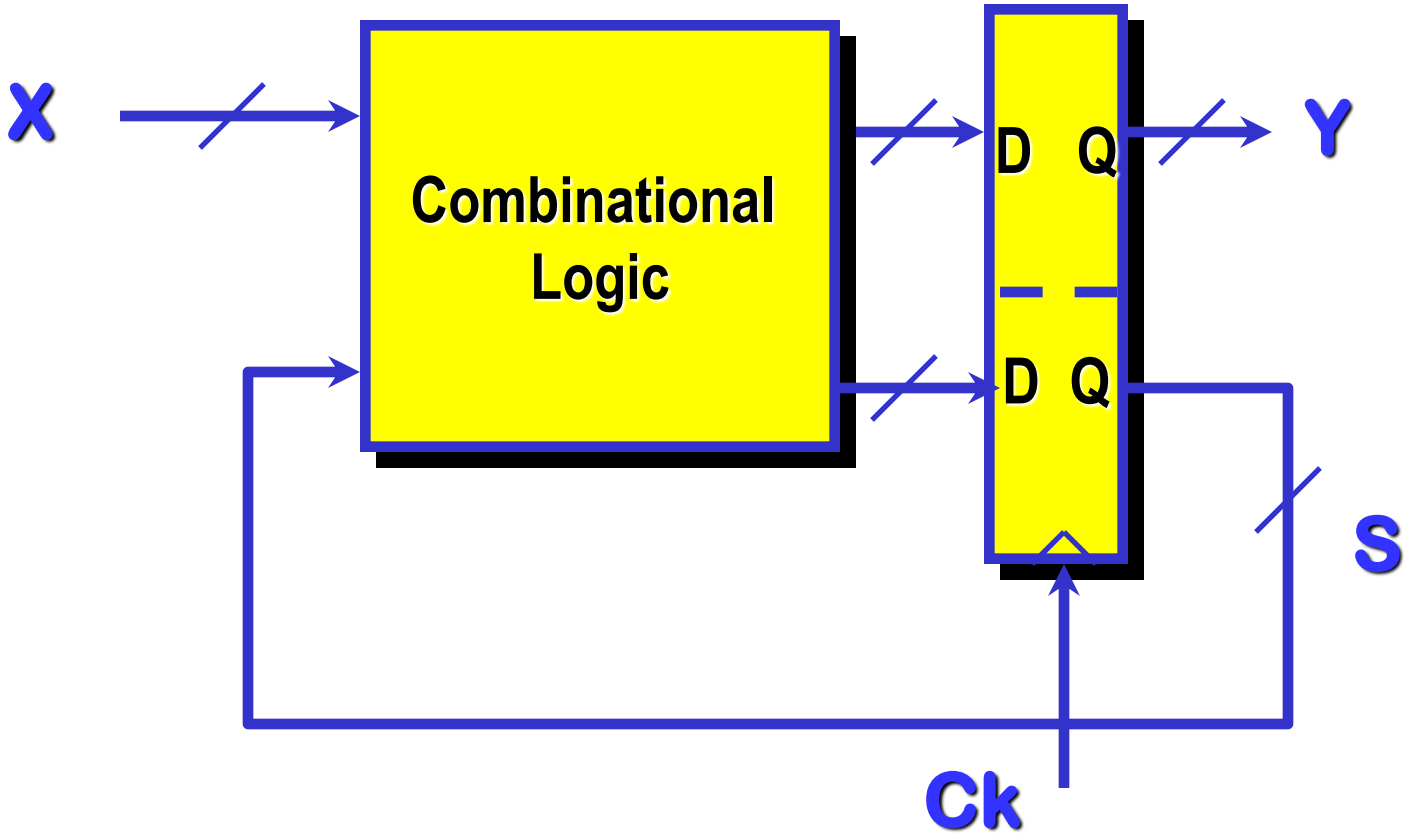


Modified MEALY machine 1/2

- Output are synchronized to solve stability problem
- Output are delayed w.r.t MEALY
- Synchronous Logic
- It is a MOORE Machine
- Sometimes required one internal state less than MOORE Machine

Modified MEALY machine 2/2

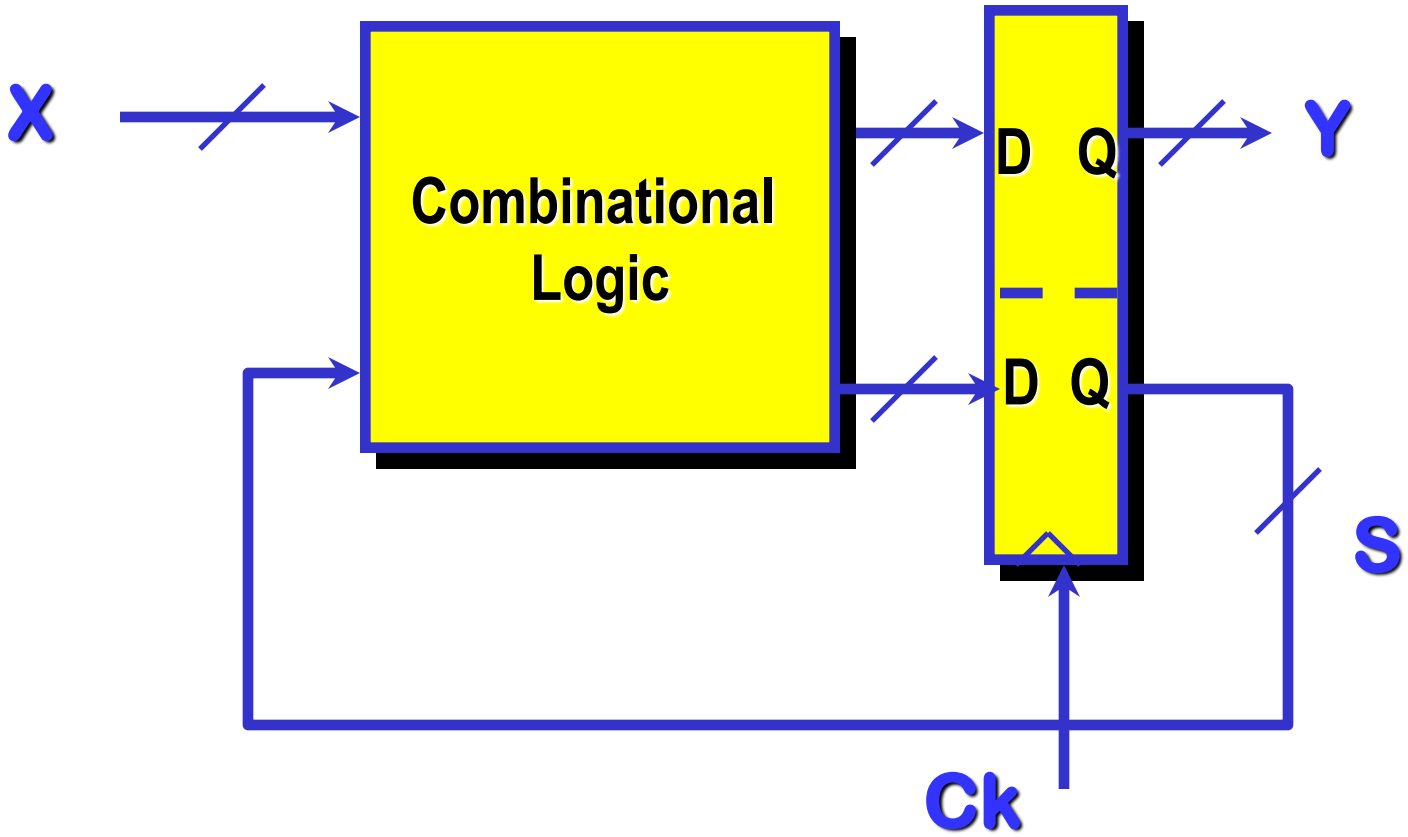
- Block Diagram



Finite State Machine: Basic Elements

Combinational Logic

Memory Element (Flip Flop D edge triggered)



Finite State Machine: Basic Elements

- How to build them ?
  - Static Logic (hard node)
  - Dynamic Logic (soft node)

Hard-Node (Static Logic)

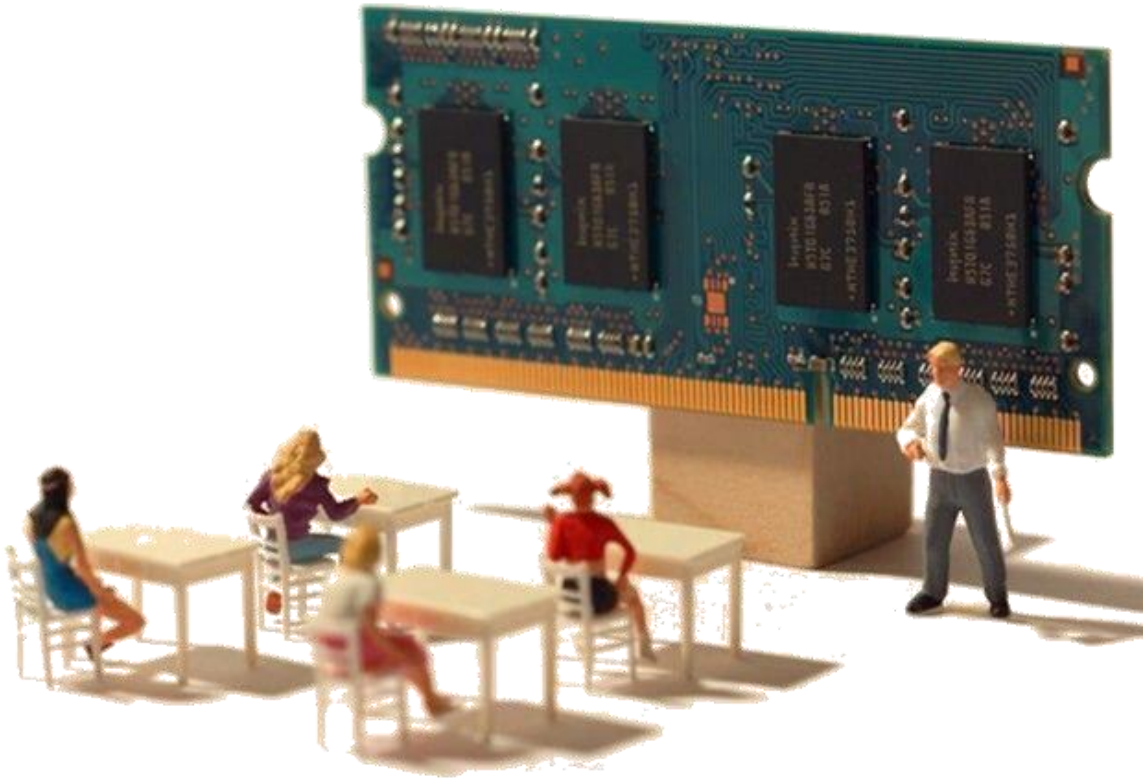
- Have positive feedback (regeneration) with an internal connection between the output and the input. Storage is performed through circuit topology
- Preserve state as long as the power is on
- Useful when updates are infrequent

Soft Node (Dynamic Logic)

- Store state on parasitic capacitors
- Only hold state for short periods of time
- Require periodic refresh
- Usually simpler, so higher speed and lower power

End Review, Questions ?

- Sync. and Async Sequential Logic
- Finite State Machine (MEALY, MOORE, MEALY Rit.)



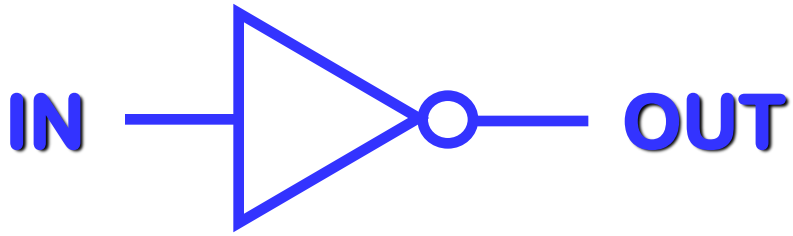
Outline

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- Example: Pass Transistor Chain
  - Dynamic Analysis
  - Area – Speed Trade-Off

INVERTER

- Implementation of the logic NOT Function

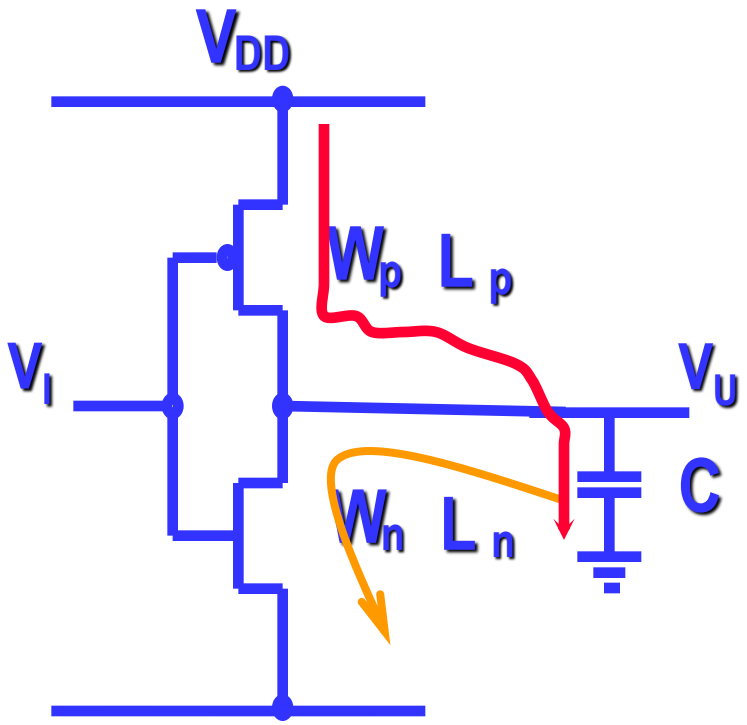
Logic Symbol



Truth Table

IN	OUT
0	1
1	0

Propagation Delay



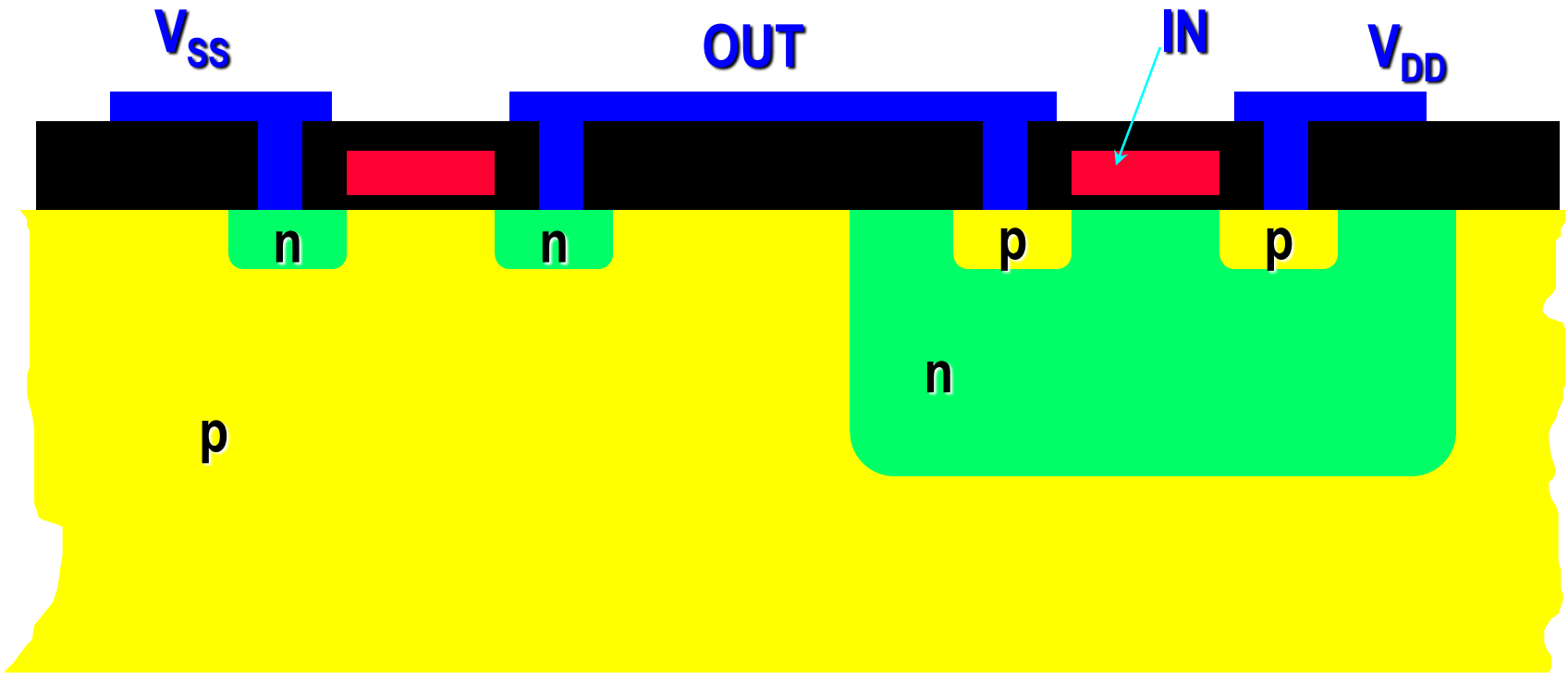
$$t_{pHL} \propto \frac{KC}{\beta_n} \frac{1}{\mathrm{V_{DD}} - V_{Tn}}$$

$$t_{pLH} \propto \frac{KC}{\beta_p} \frac{1}{\mathrm{V_{DD}} + V_{Tp}}$$

$$\beta_n = \mu_n C_{ox} \frac{W_n}{L_n} \qquad \beta_p = \mu_p C_{ox} \frac{W_p}{L_p}$$

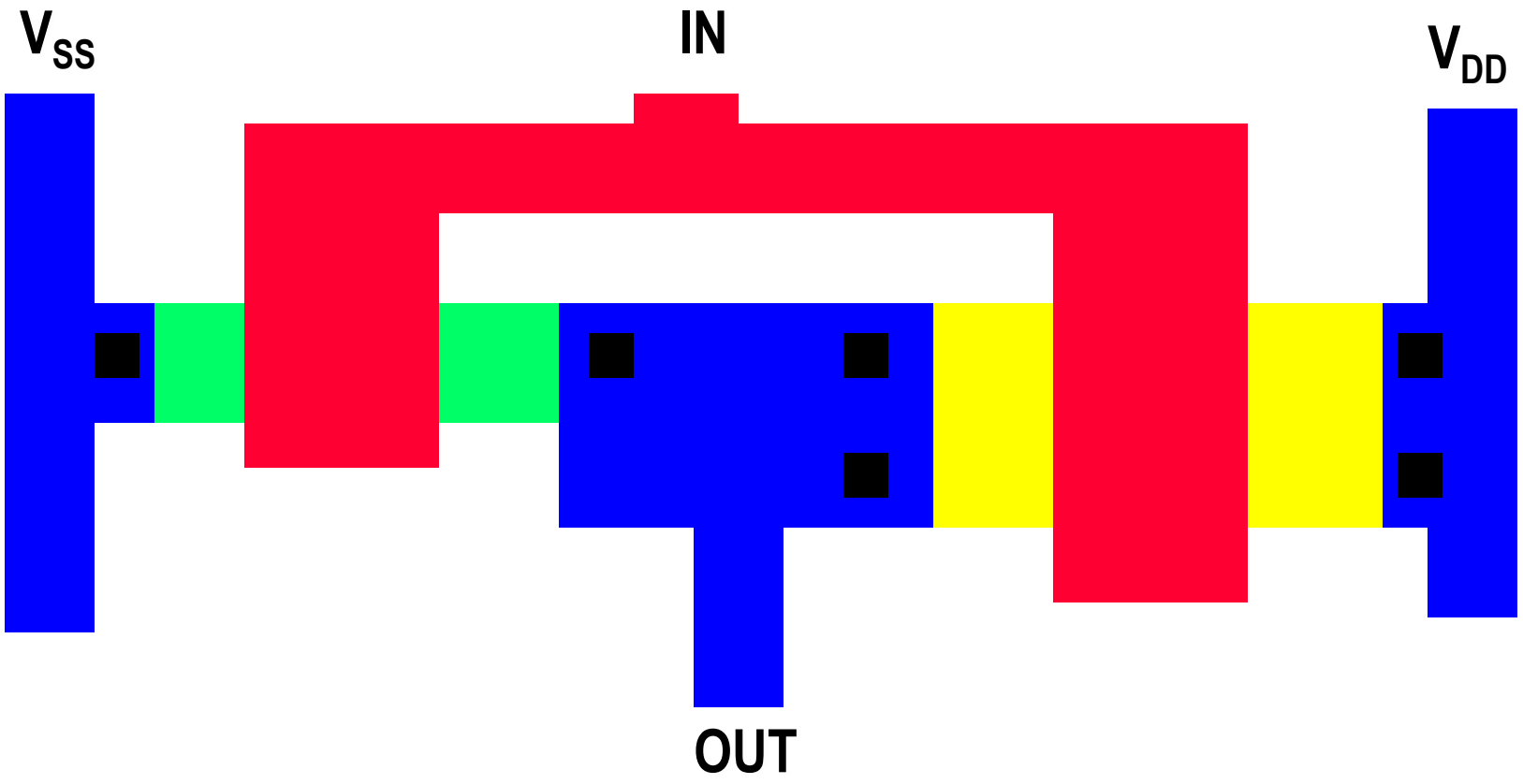
INVERTER Lay-out 1

- Cross Section for "n" well process



INVERTER Lay-out 2

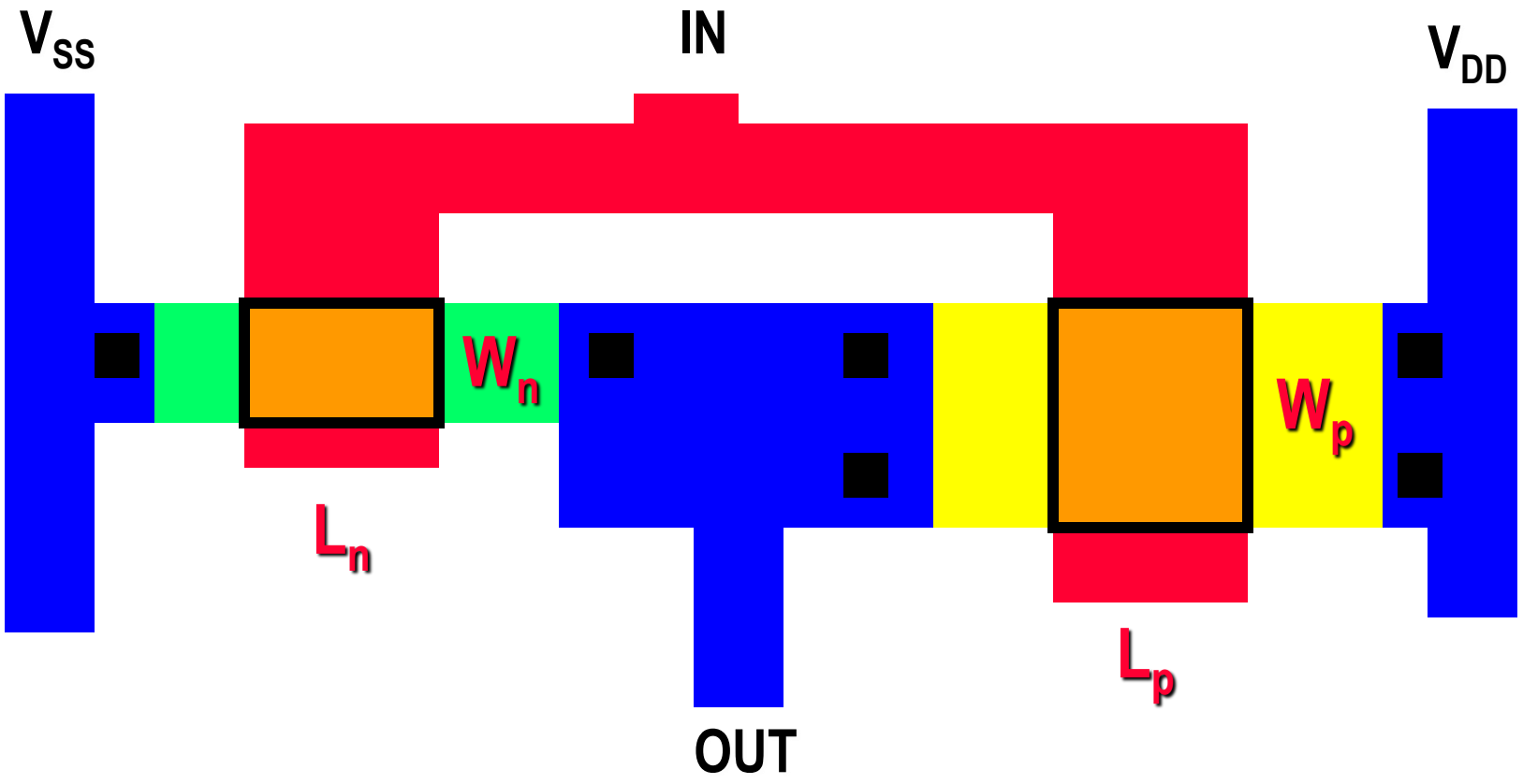
✧ Top view



INVERTER Lay-out 2

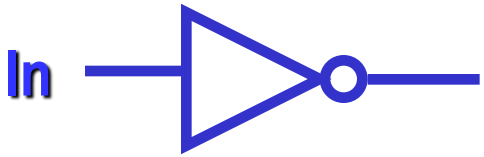
✧ Top view

$A \sim W_n L_n + W_p L_p$

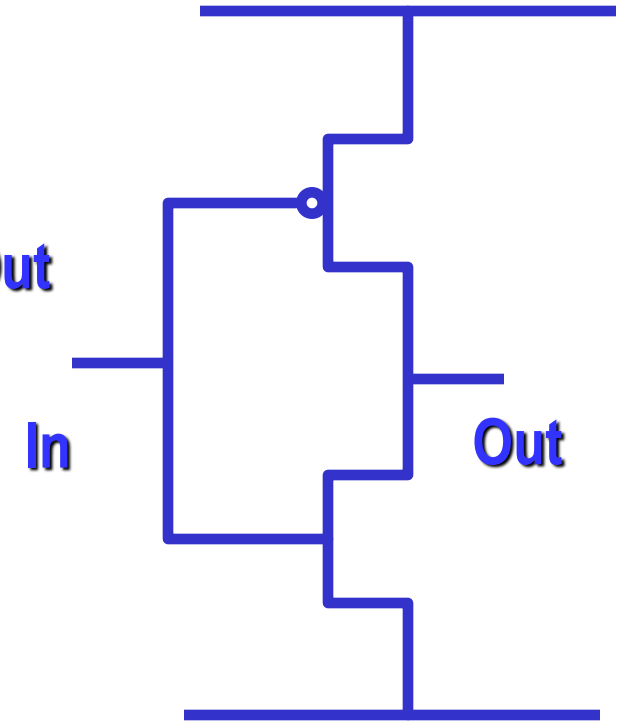


Stick Diagram

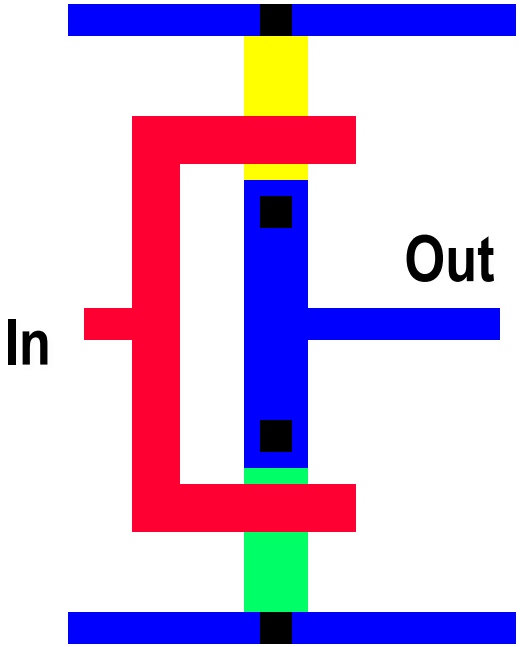
✧Logic Symbol



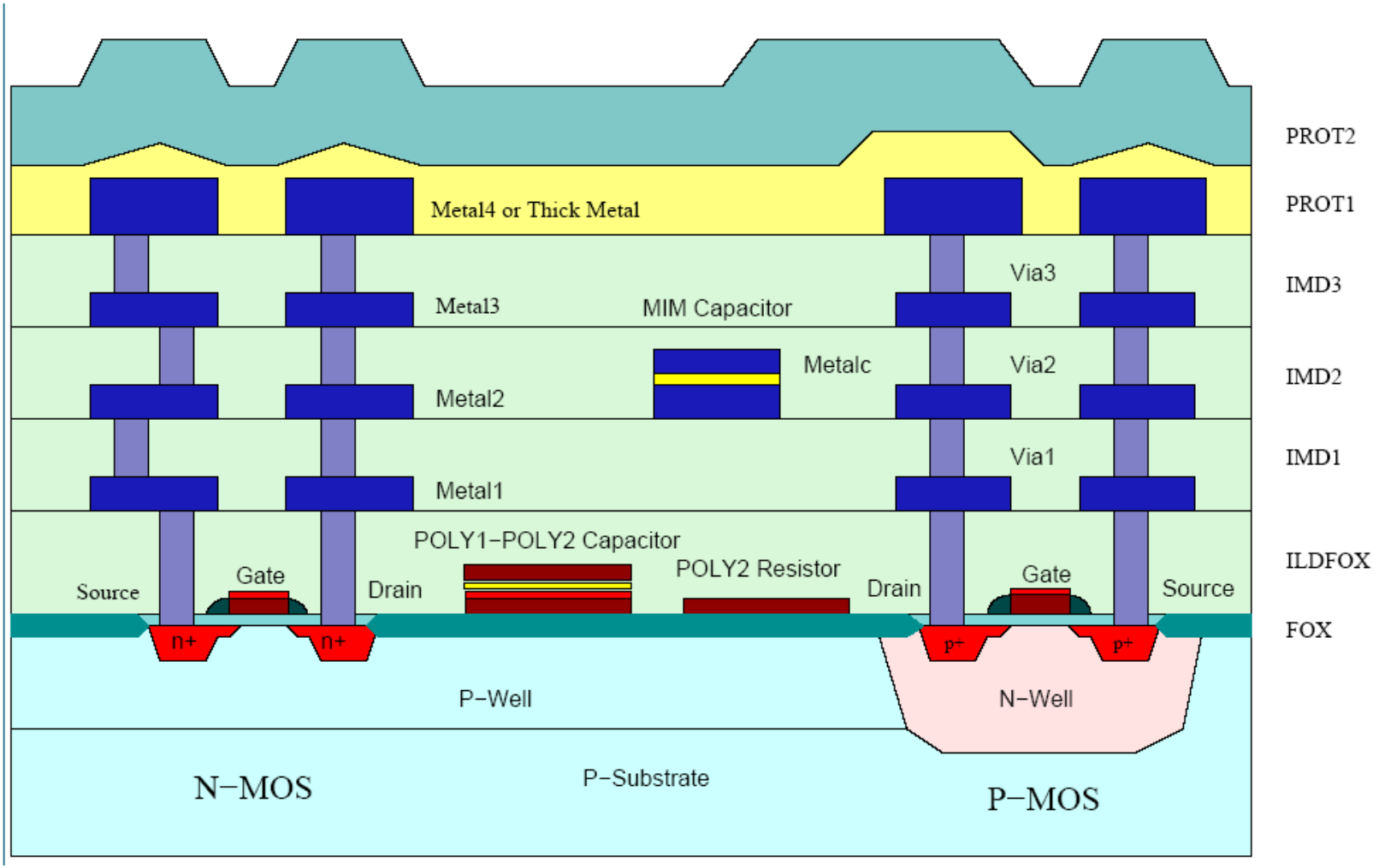
Circuit



Lay-out



Wafer Cross-Section Austriamicrosystems, 0,35 um CMOS



Austriamicrosystems, 0,35 um CMOS Technology Overview

Process technology specifications	units	C35B3C3	C35B4C3
Drawn MOS Channel Length	µm	0,35	
Operating Voltage	V	2.5 - 3.6	IO -5.5
Number of Masks	#	14	20
Number of Masking layers	#	19	24
Number of Metal Layers	#	3	4
Number of Poly Layers	#	2	
Substrate Type		p	
Diffusion Pitch	µm	0,9	
Metal1/2/3/4 Pitch	µm	0.95 / 1.1 / 1.1 / 1.2	
Metal1/2/3/4 conacted Pitch	µm	1.05 / 1.2 / 1.2 / 1.3	
Poly1 Pitch	µm	0,8	
High Resistive Poly	kOhm/#	-	1,2
Poly1/Poly2 Precision Caps	fF/µm²	0,9	
N/PMOS Channel Length	µm	0.30/0.30	
N/PMOS Saturation Current	µA/µm	520 / 240	

Standard-Cell Austriamicrosystems, 0,35 um CMOS Technology Library



0.35µm CMOS

Digital Standard Cell Databook

Inverter

INVX1

www.ams.com

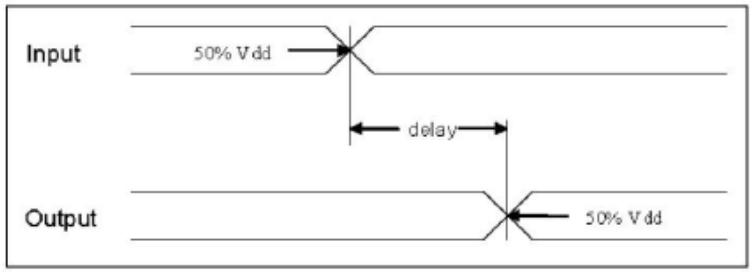
Conditions for characterization library **c35_CORELIBD_BC**, corner **c35_CORELIBD_BC_best**: Vdd= **3.63**V, Tj= **-50.0** deg. C .
Output transition is defined from **20%** to **80%** (rising) and from **80%** to **20%** (falling) output voltage.
Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	29.120 um^2
Equation	Q = "!A"
Type	Combinational
Input	A
Output	Q

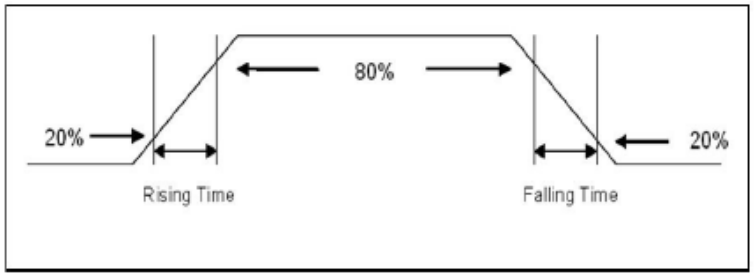


State Table	
A	Q
L	H
H	L

Capacitance [fF]	
A	2.8210



Propagation Delay



Transition Time

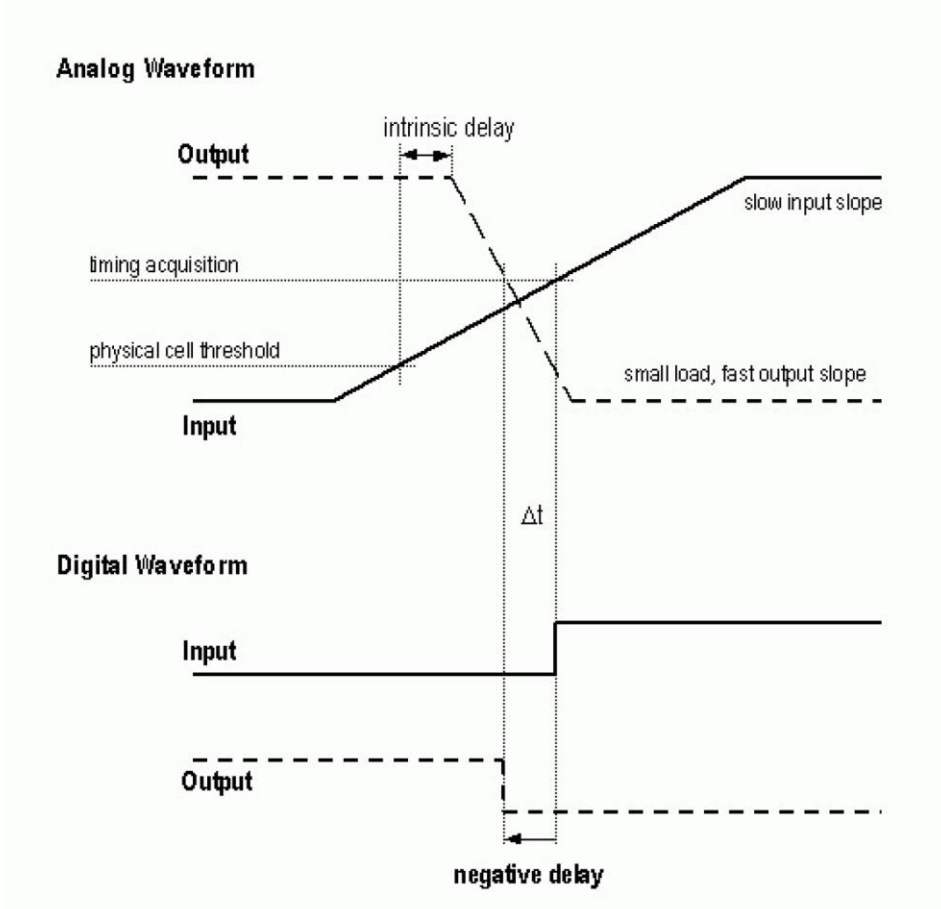
Interpreting « Negative Delays »

www.ams.com

Interpreting "Negative Delays"

Under certain conditions, delay calculations may result in negative delay values.

These “negative delays” are mainly caused by the difference between timing acquisition and the physical cell threshold as shown in the drawing below.



Inverter

INVX1

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Conditions for characterization library **c35_CORELIBD_BC**, corner **c35_CORELIBD_BC_best**: Vdd= **3.63**V, Tj= **-50.0** deg. C .
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Strength	1
Cell Area	29.120 um^2
Equation	Q = "!A"
Type	Combinational
Input	A
Output	Q



State Table	
A	Q
L	H
H	L

Capacitance [fF]	
A	2.8210

Propagation Delay [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		5.00	100.00	5.00	100.00
A to Q	fall	0.04	0.41	-0.36	0.63
	rise	0.06	0.67	0.83	1.88

Output Transition [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		5.00	100.00	5.00	100.00
A to Q	fall	0.04	0.54	0.70	1.43
	rise	0.08	1.03	0.62	1.58

Inverter

INVX1

www.ams.com

Conditions for characterization library **c35_CORELIBD_BC**, corner **c35_CORELIBD_BC_best**: Vdd= **3.63**V, Tj= **-50.0** deg. C .
Output transition is defined from **20%** to **80%** (rising) and from **80%** to **20%** (falling) output voltage.
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Strength	1
Cell Area	29.120 um^2
Equation	Q = "!A"
Type	Combinational
Input	A
Output	Q



State Table	
A	Q
L	H
H	L

Capacitance [fF]	
A	2.8210

Dynamic Power Consumption [nW/MHz]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		5.00	100.00	5.00	100.00
A to Q	fall	1.93	2.35	516.63	510.09
	rise	38.82	40.07	813.89	712.95

Leakage [pW]
0.26

Inverter

INVX1

www.ams.com

Conditions for characterization library **c35_CORELIBD_BC**, corner **c35_CORELIBD_BC_best**: Vdd= **3.63**V, Tj= **-50.0** deg. C .
Output transition is defined from **20%** to **80%** (rising) and from **80%** to **20%** (falling) output voltage.
Propagation delay is measured from **50%** (input rise) or **50%** (input fall) to **50%** (output rise) or **50%** (output fall).

Strength	1
Cell Area	29.120 um^2
Equation	Q = "!A"
Type	Combinational
Input	A
Output	Q



State Table	
A	Q
L	H
H	L

Capacitance [fF]	
A	2.8210

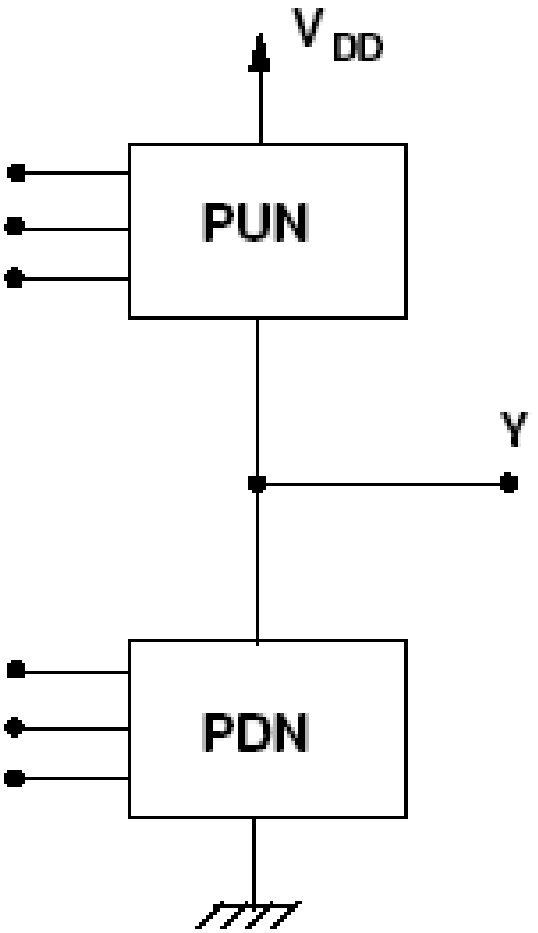
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Leakage [pW]
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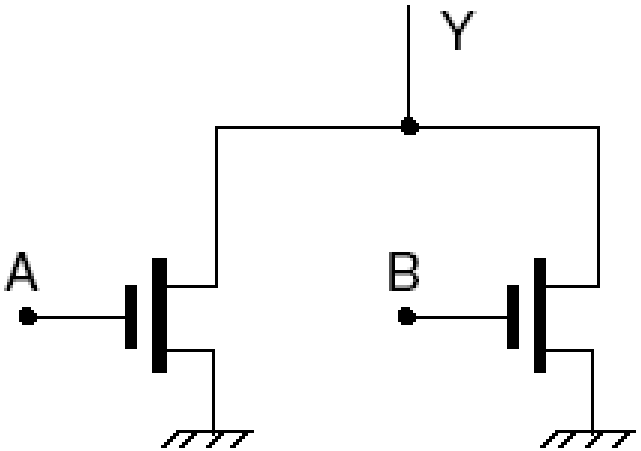
CMOS Technology Corners

	BEST	TYPICAL	WORST
Temperature	-40 - 0 °C	25 °C	85 - 125 °C
Power Supply	1,1*VDD	VDD	0,9*VDD
Process	FAST	TYP	SLOW

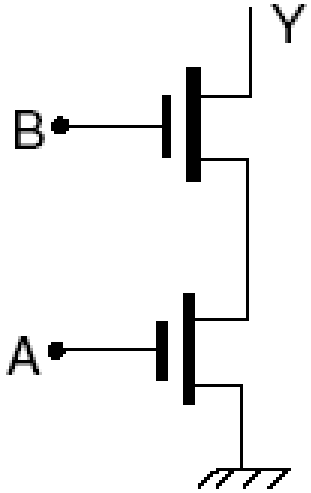
Complementary CMOS



PDN



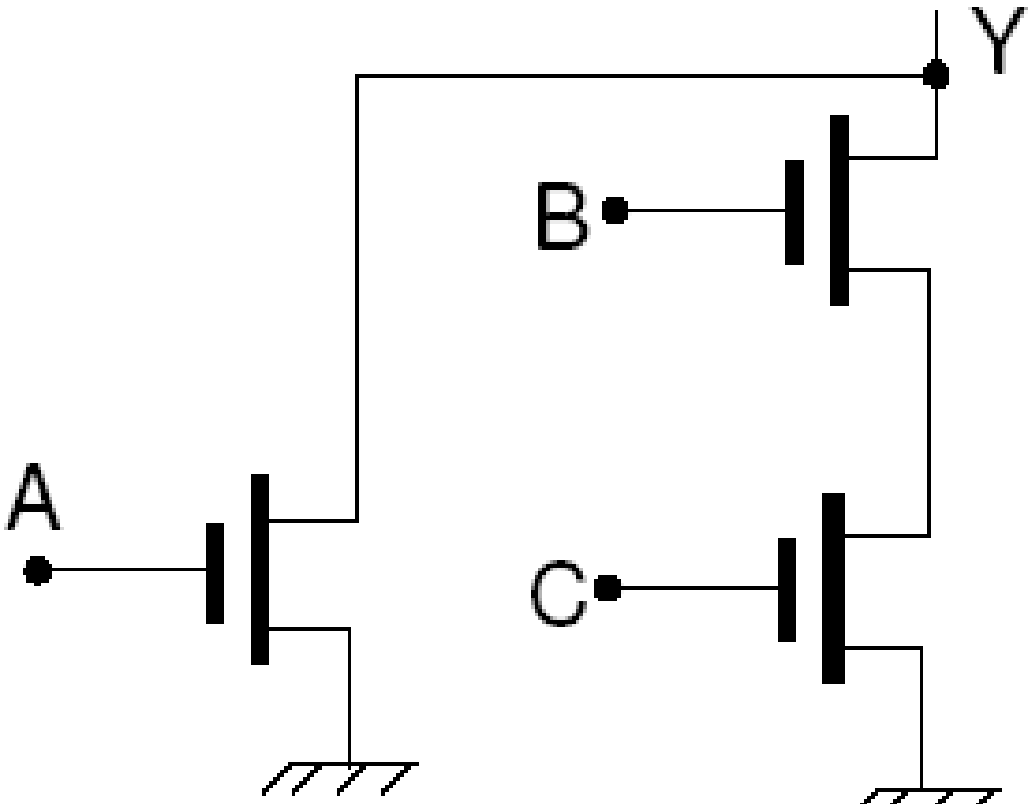
$\overline{Y}=A+B$



$\overline{Y}=AB$

A	B	OR	AND
0	0	0	0
0	1	1	0
1	0	1	0
1	1	1	1

PDN

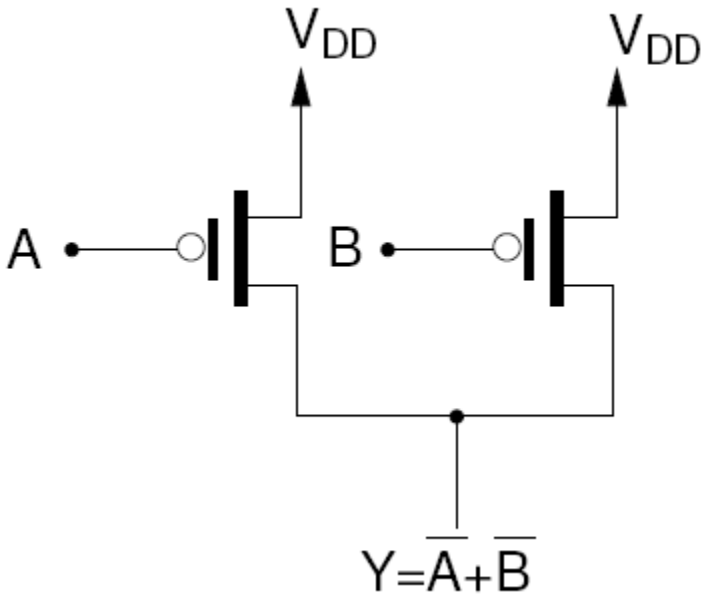


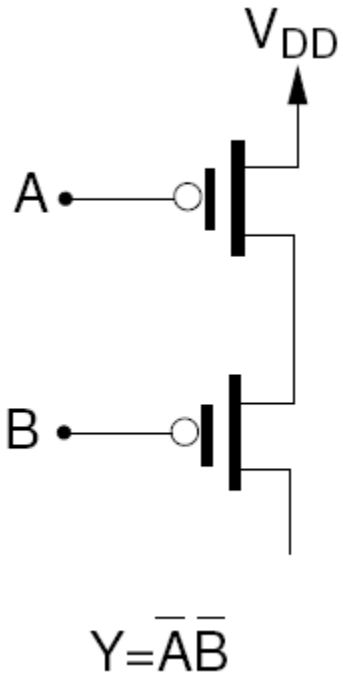
$\overline{Y}=A+BC$

PUN

- PUN is DUAL of PDN
- $\overline{A+B} = \overline{A} \bullet \overline{B}$
- $\overline{A \bullet B} = \overline{A} + \overline{B}$

A	B	NAND
0	0	1
0	1	1
1	0	1
1	1	0



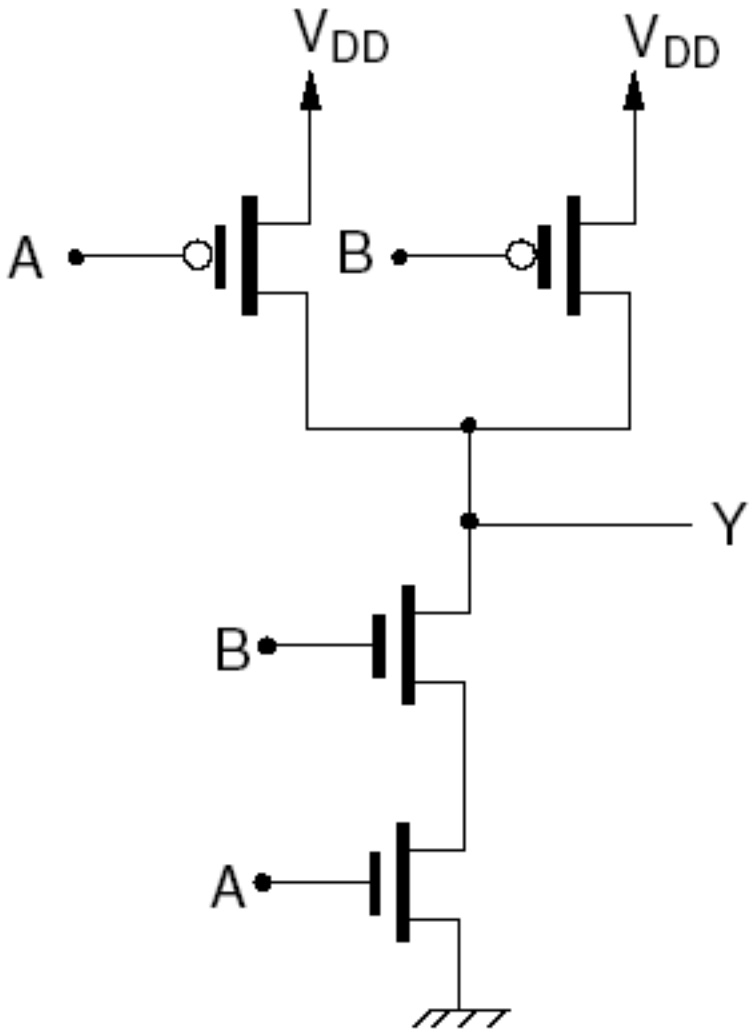


PDN / PUN

- $\overline{Y} = f_{PDN}(A, B)$
- $Y = f_{PUN}(\overline{A}, \overline{B})$

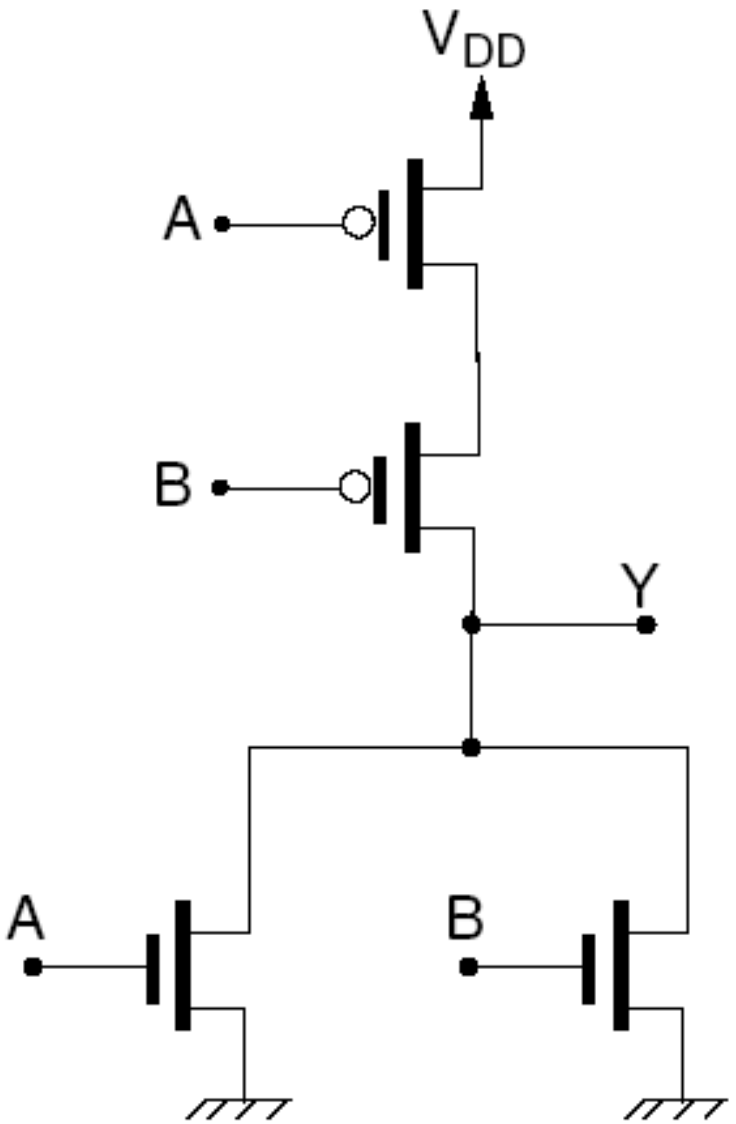
NAND 2

- $Y = \overline{A \bullet B}$
- PUN -> $Y = \overline{A} + \overline{B}$
- PDN -> $\overline{Y} = A \bullet B$



NOR 2

- $Y = \overline{A + B}$
- PUN -> $Y = \overline{A} \bullet \overline{B}$
- PDN -> $\overline{Y} = A + B$



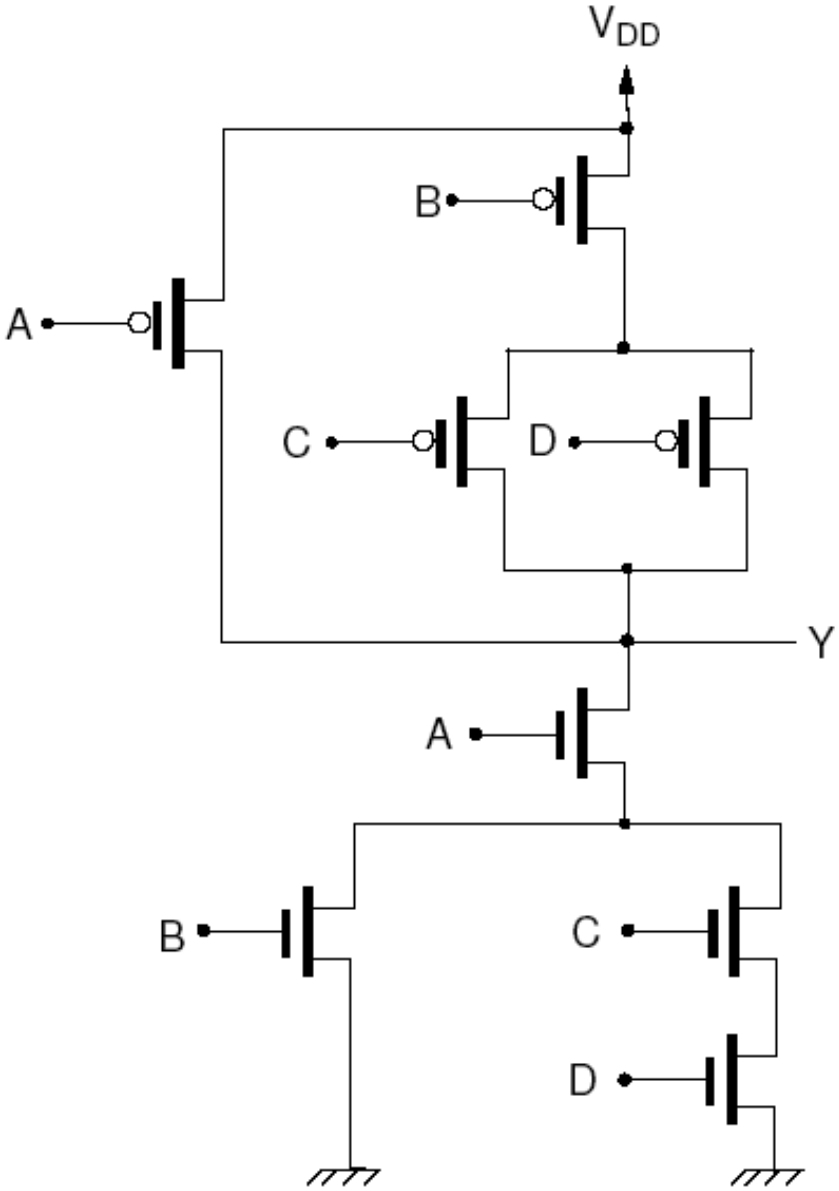
COMPLEX FUNCTION

$$Y = \overline{A(B + C\dot{D})}$$

PUN ->

$$\begin{aligned} Y &= \overline{A(B + CD)} \\ &= \overline{A} + \overline{B + CD} \\ &= \overline{A} + \overline{B}\,\overline{CD} \\ &= \overline{A} + \overline{B}(\overline{C} + \overline{D}) \end{aligned}$$

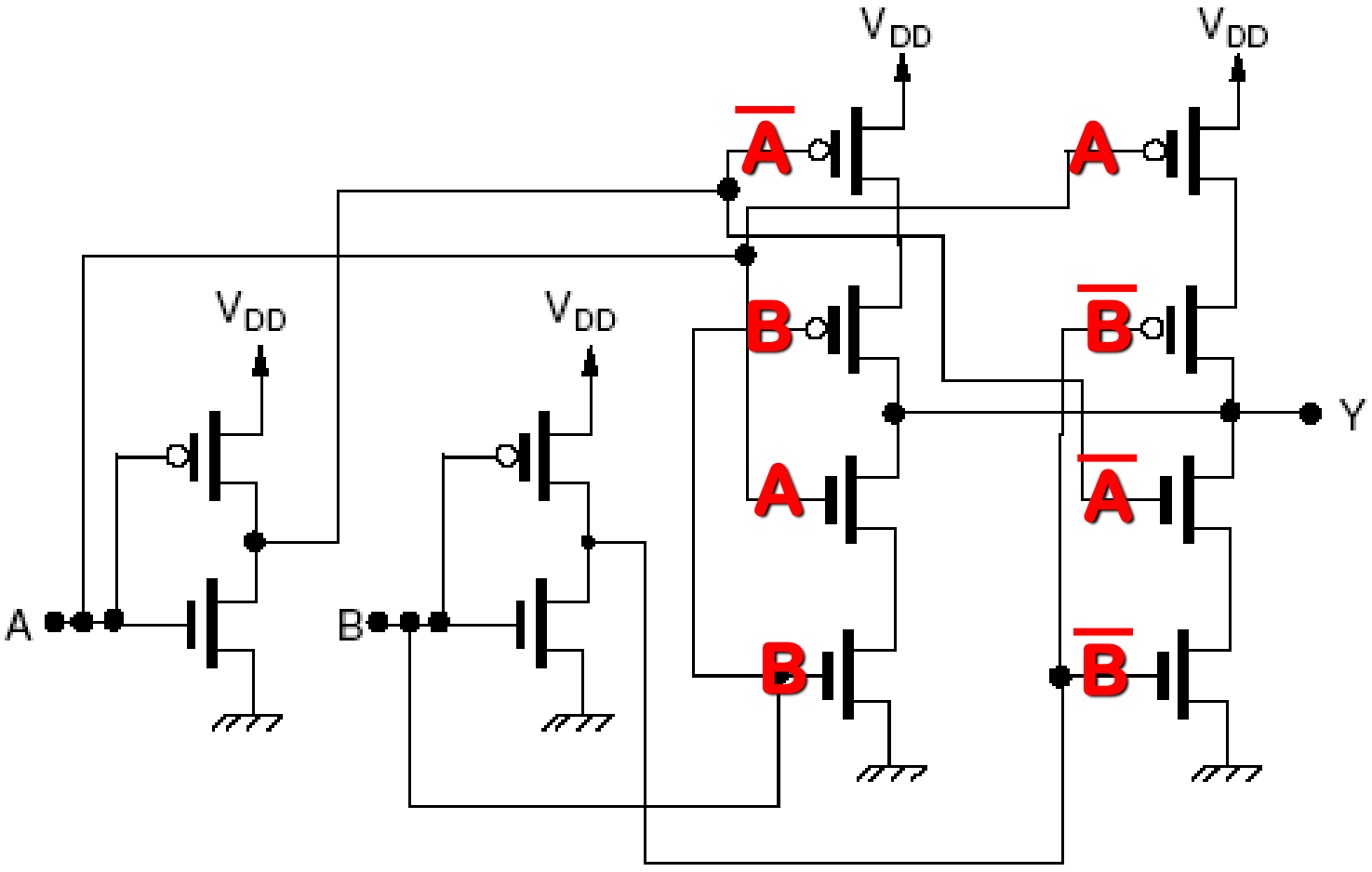
PDN -> $\overline{Y} = A(B + CD)$



A	B	XOR
0	0	0
0	1	1
1	0	1
1	1	0

XOR

$$Y = A\overline{B} + \overline{A}B.$$



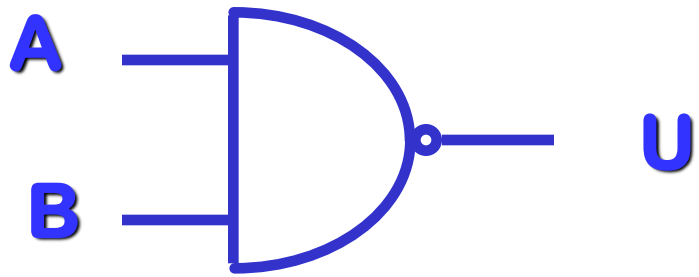
NAND

Truth Table

A	B	U
0	0	1
0	1	1
1	0	1
1	1	0

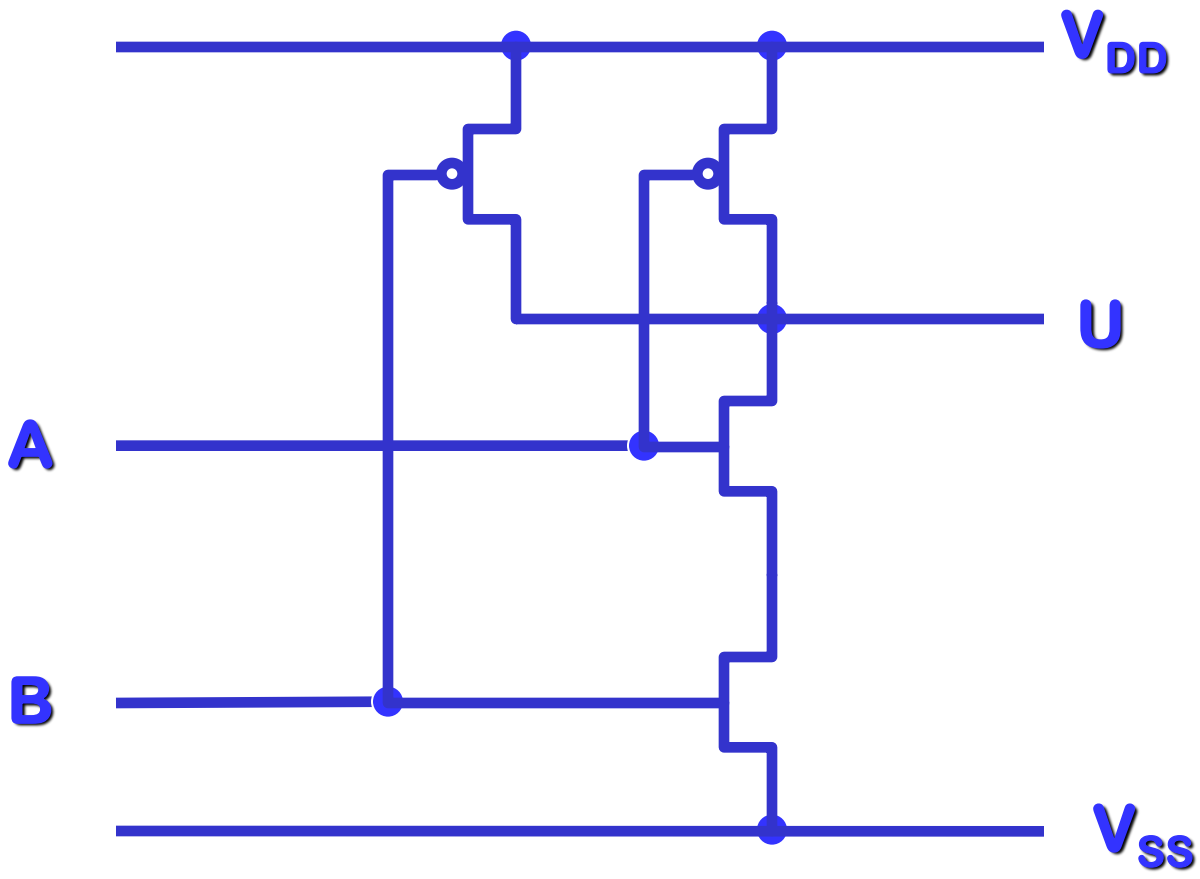
Logic Symbol

$$U = \overline{A * B}$$



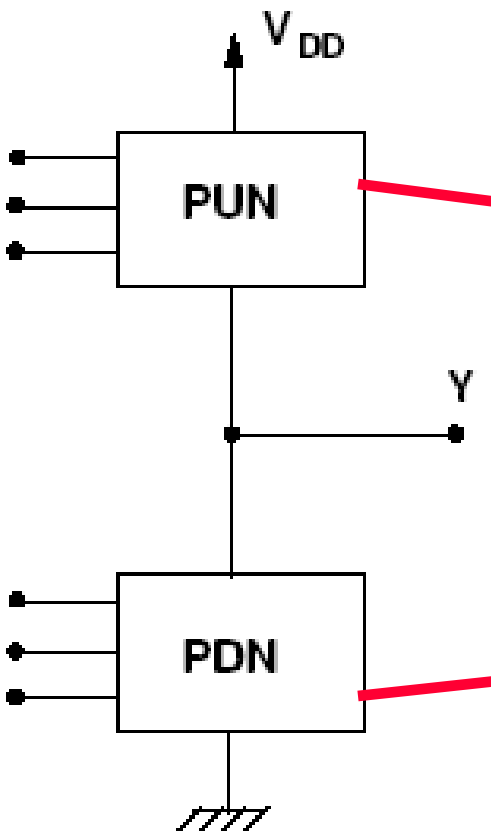
CMOS Implementation

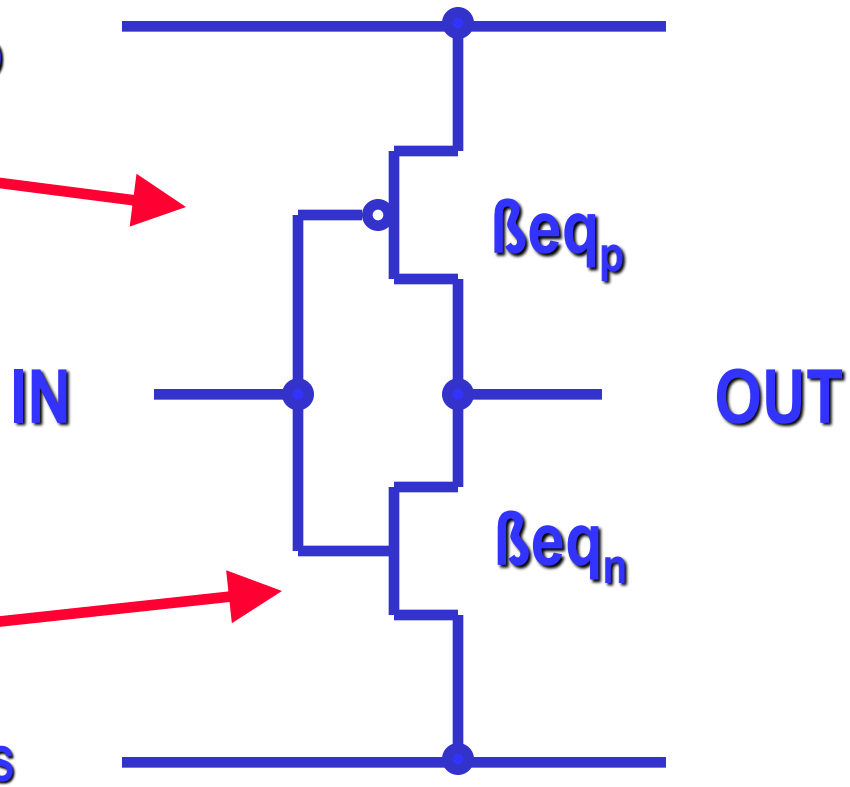
- Electric Circuit



Dynamic A

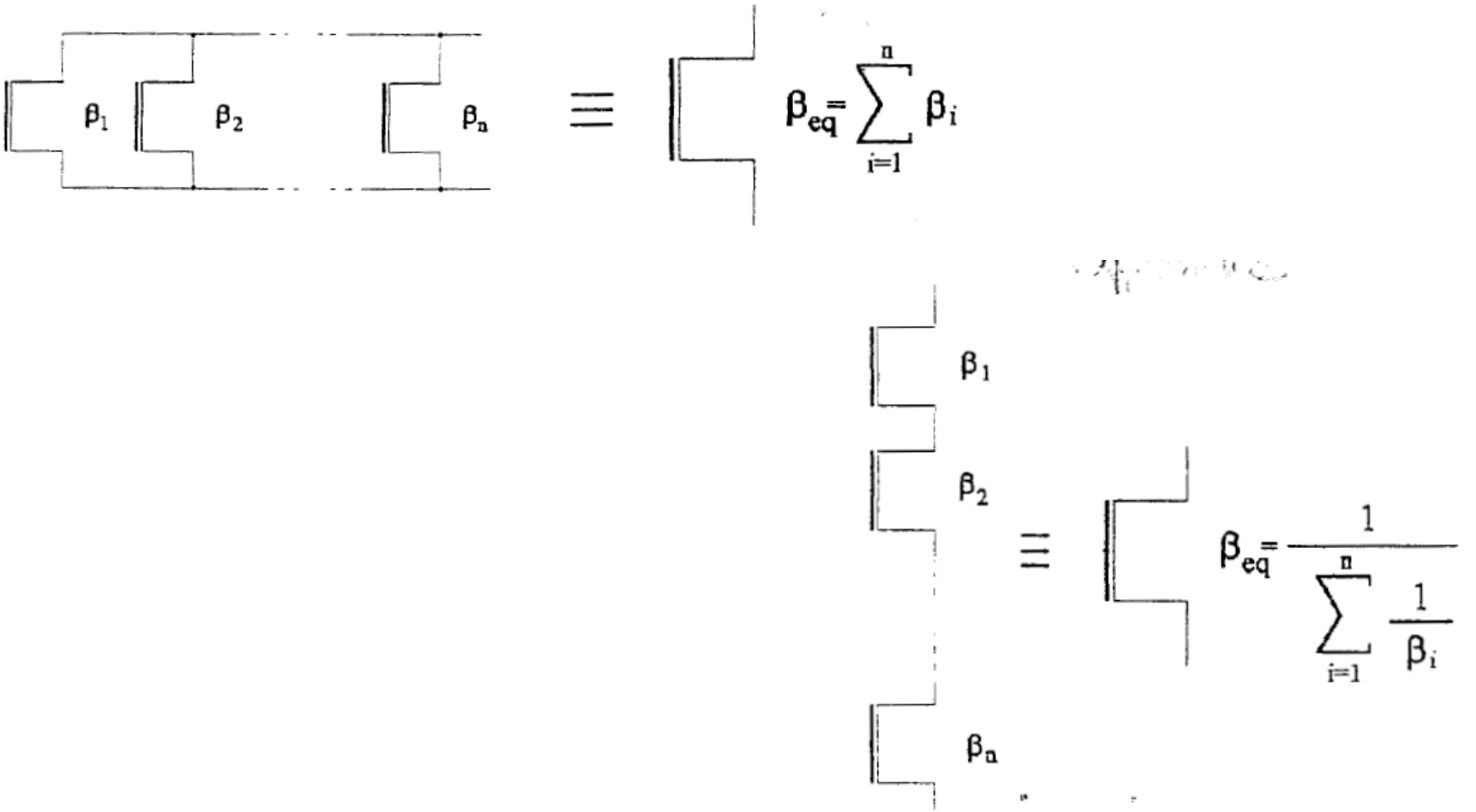
$$t_{pHL} \propto \frac{KC}{\beta_n} \frac{1}{V_{DD} - V_{Tn}}$$



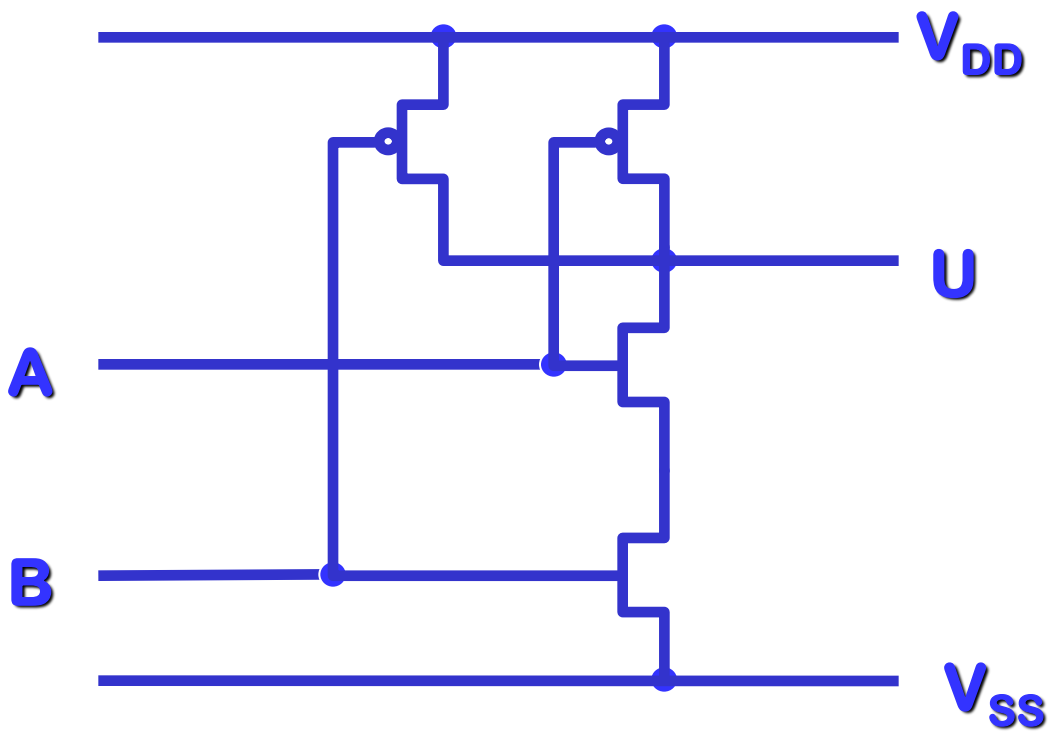


$$t_{pLH} \propto \frac{KC}{\beta_p} \frac{1}{V_{DD} + V_{Tp}}$$

Dynamic Analysis



Dynamic Analysis



A	B	U	
0	0	1	$2\,ß_p$
0	1	1	$ß_p$
1	0	1	$ß_p$
1	1	0	$ß_n/2$

$$t_{pLH} \propto \frac{KC}{\beta_{eqp}} = \frac{KC}{\beta_p}$$

$$t_{pHL} \propto \frac{KC}{\beta_{eqn}} = \frac{2KC}{\beta_n}$$

Dynamic Analysis

$$t_{pLH} \propto \frac{KC}{\beta_{eqp}} = \frac{KC}{\beta_p} \qquad t_{pHL} \propto \frac{KC}{\beta_{eqn}} = \frac{2KC}{\beta_n}$$

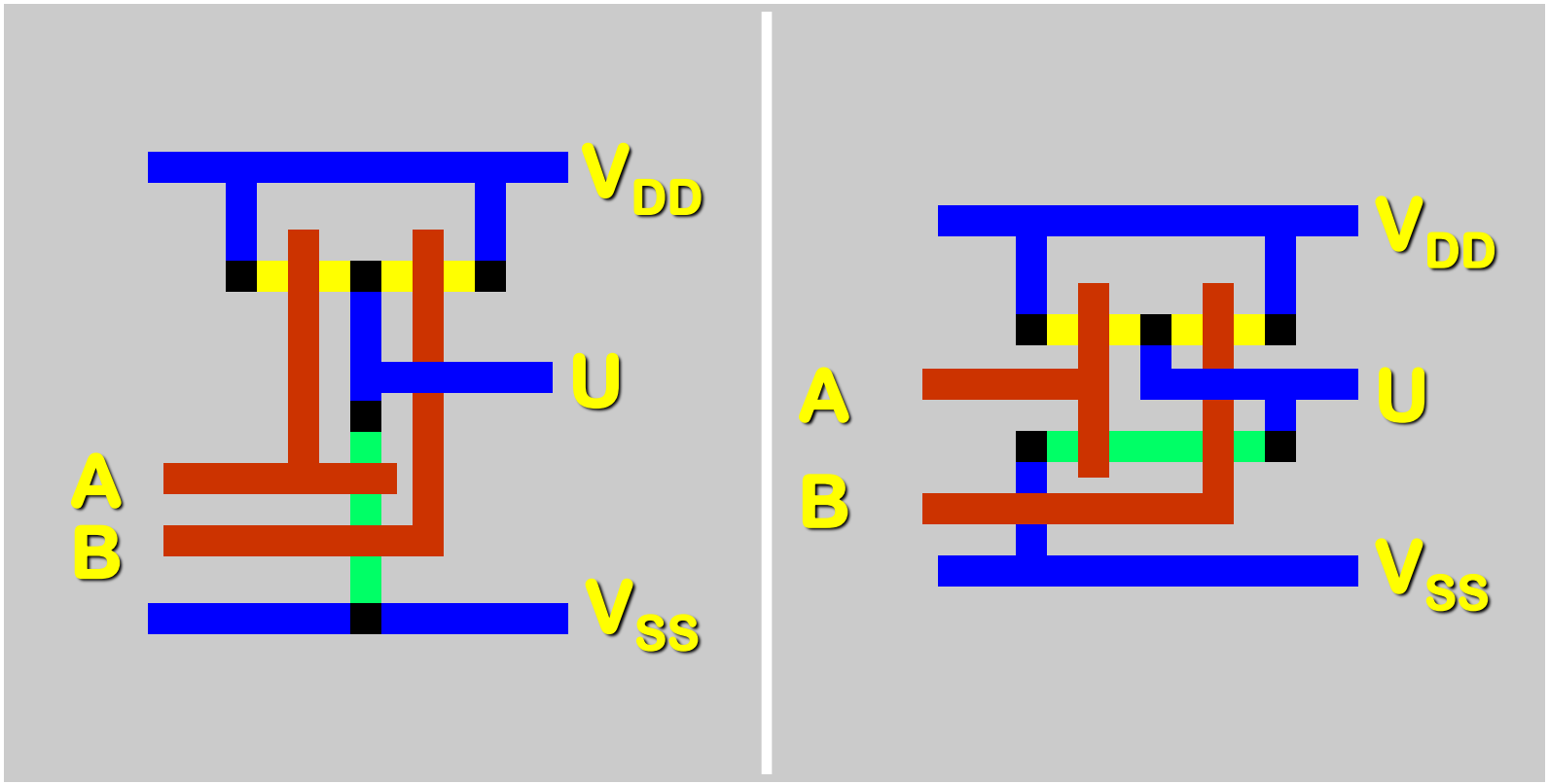
✧ $tp_{HL} = tp_{LH}$ in the worst case

$$\beta_p = \frac{\beta_n}{2} \rightarrow \beta_n = 2\beta_p \rightarrow \mu_n \frac{W_n}{L_n} = 2\mu_p \frac{W_p}{L_p}$$

If $\mu_p = \frac{\mu_n}{2}$ $L_n = L_p = L_{\min}$ ⟶ $W_n = W_p$

Stick Diagram

- Different topological solutions



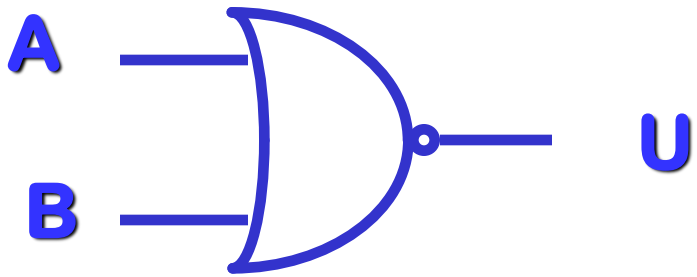
NOR

Truth Table

A	B	U
0	0	1
0	1	0
1	0	0
1	1	0

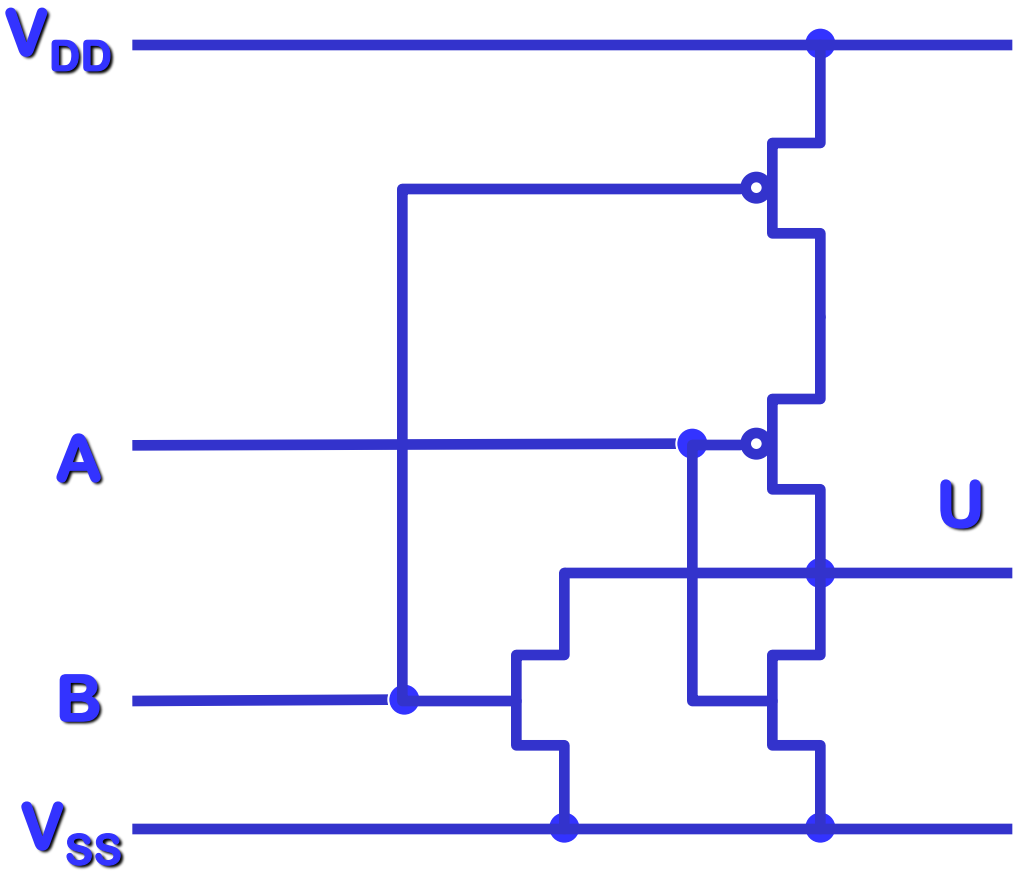
Logic Symbol

$$U = \overline{A + B}$$



Dynamic Analysis

✧ Electric Circuit

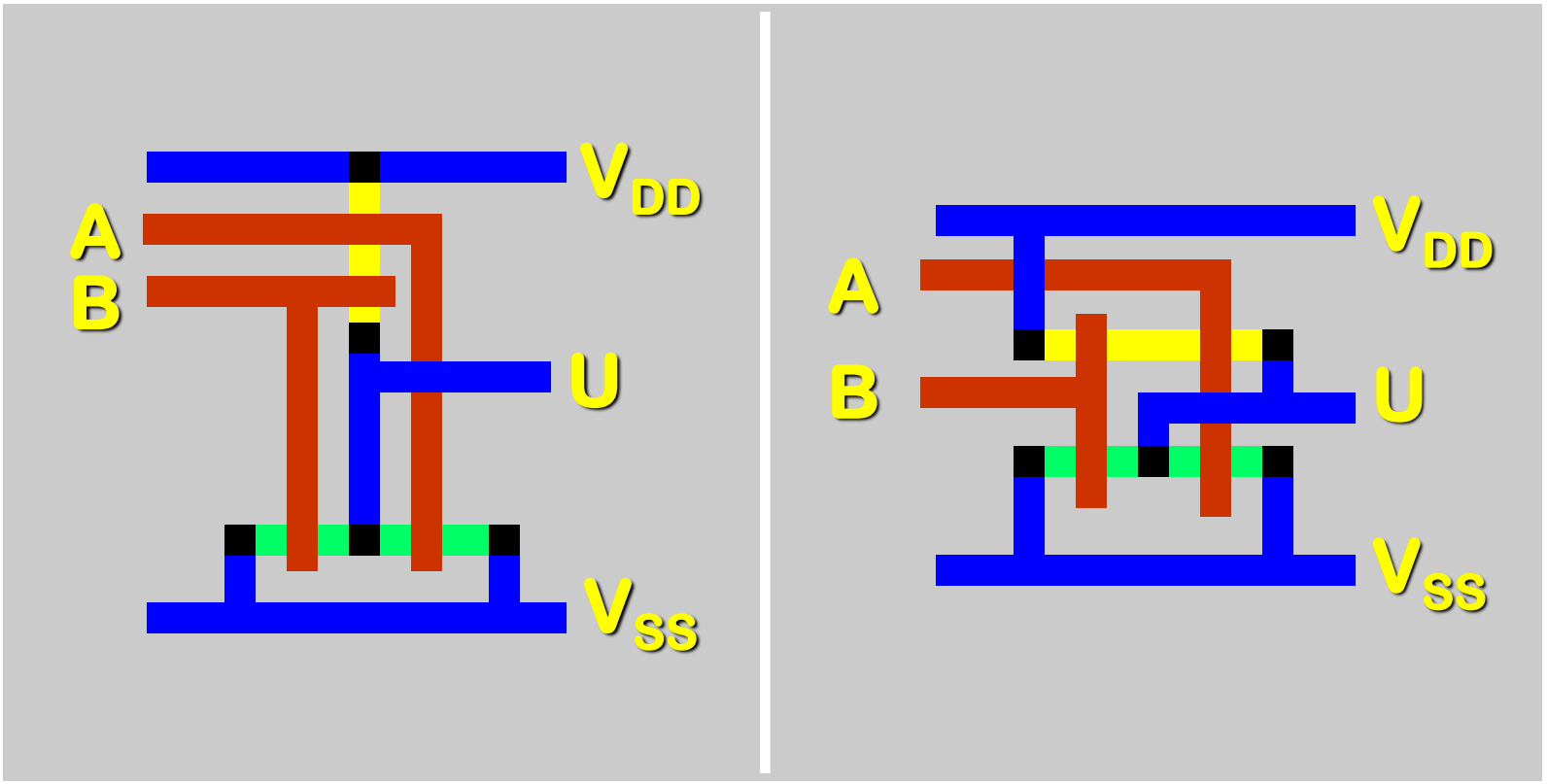


A	B	U	
0	0	1	$ß_p/2$
0	1	0	$ß_n$
1	0	0	$ß_n$
1	1	0	$2\,ß_n$

If $\mu_p = \frac{\mu_n}{2}$ $L_n = L_p = L_{\min}$ ⟶ $W_p = 4W_n$

Stick Diagram

- Different topological solutions



NOR2 – NAND2 Comparison

$$\beta eq_{NAND} = \beta_p^{NAND} = \frac{\beta_n^{NAND}}{2}$$

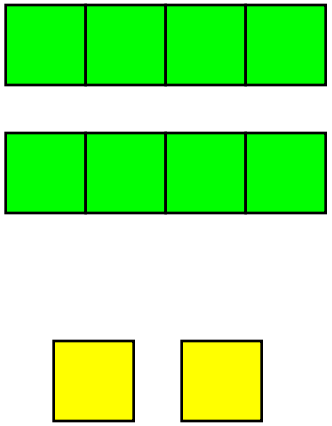
$$\beta eq_{NOR} = \beta_n^{NOR} = \frac{\beta_p^{NOR}}{2}$$

$$W_n^{NOR} = W \qquad W_p^{NOR} = 4W$$

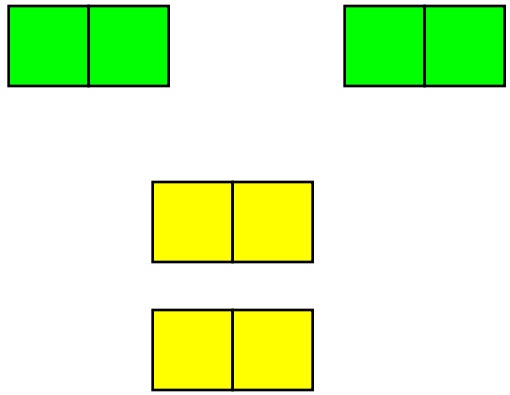
$$\beta_n^{NAND} = 2\beta_n^{NOR} \Rightarrow W_n^{NAND} = 2W_n^{NOR} = 2W$$

$$\beta_p^{NAND} = \frac{\beta_p^{NOR}}{2} \Rightarrow W_p^{NAND} = \frac{W_p^{NOR}}{2} = \frac{4W}{2W} = 2W$$

NOR2 – NAND2 Comparison



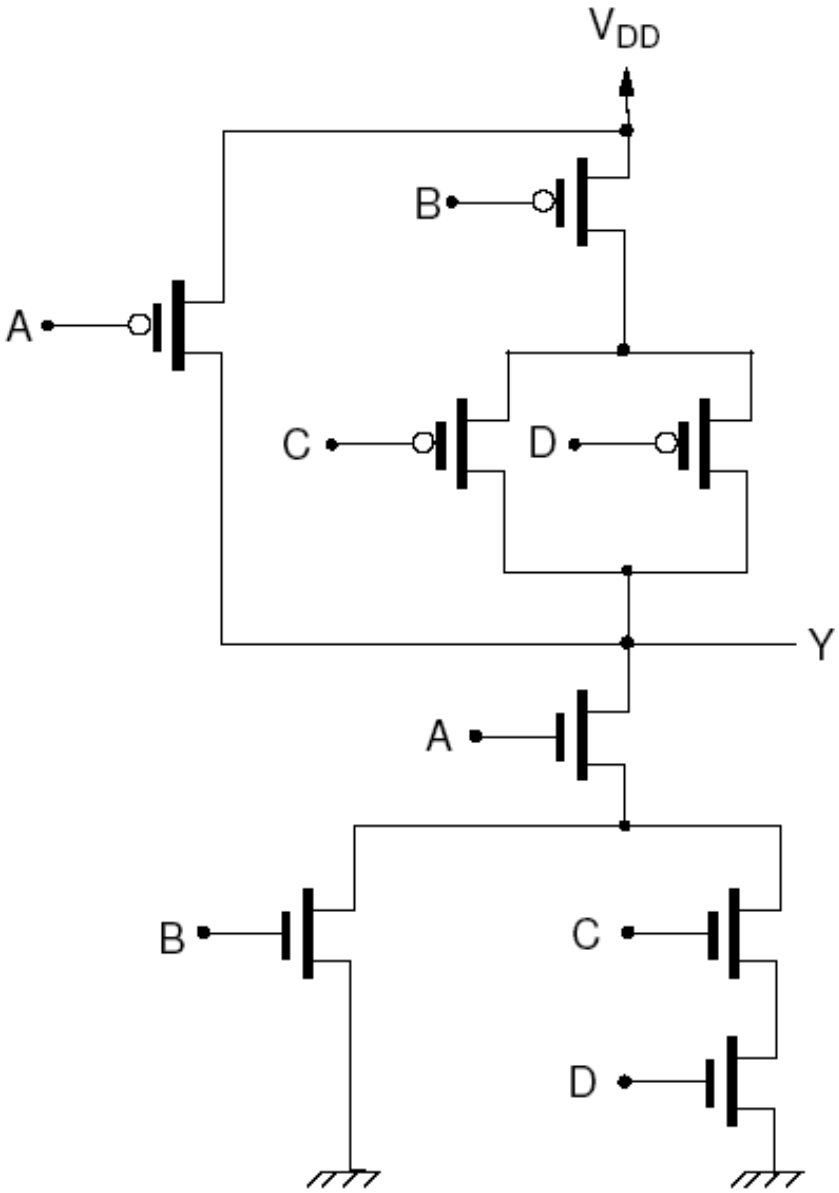
10 WL



8 WL

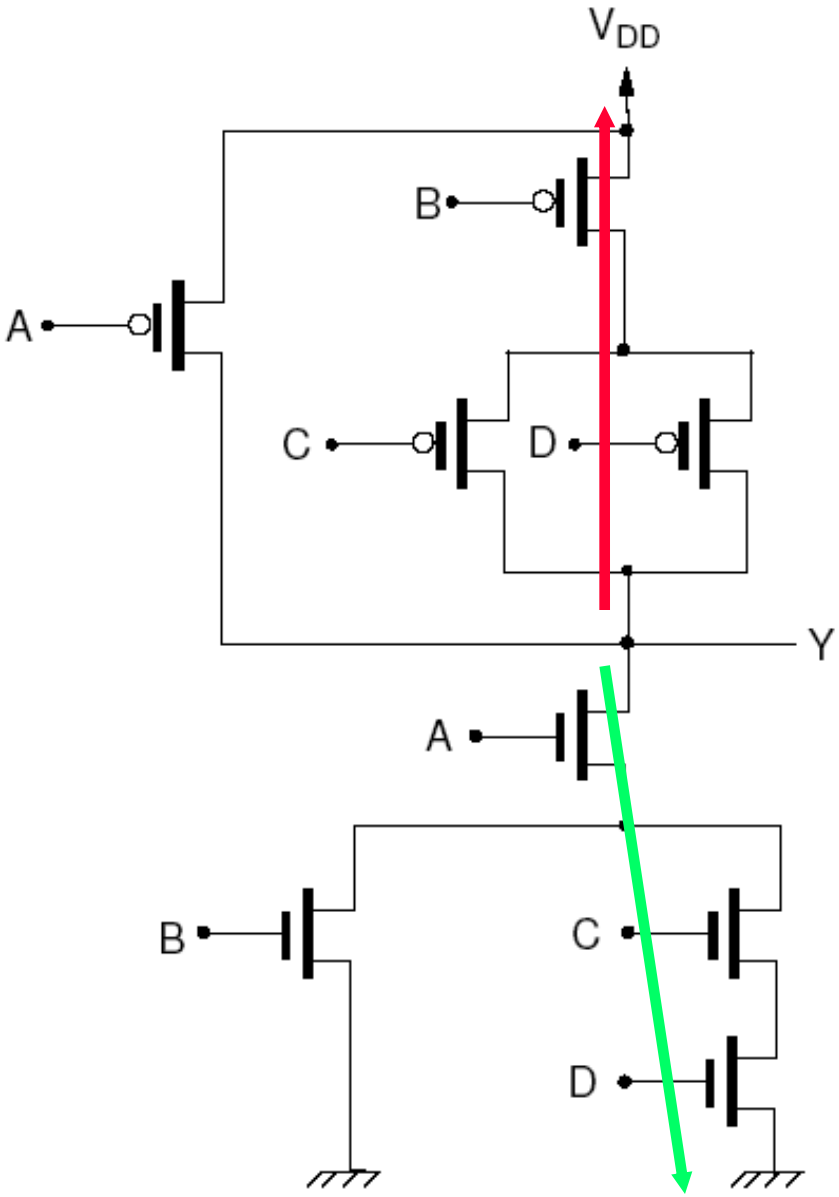
$$Y = \overline{A(B + C\dot{D})}$$

$$Y = \overline{A(B + C\dot{D})}$$



A	B	C	D	Y	ß
0	0	0	0	1	5/3
0	0	0	1	1	3/2
0	0	1	0	1	3/2
0	0	1	1	1	1
0	1	0	0	1	1
0	1	0	1	1	1
0	1	1	0	1	1
0	1	1	1	1	1
1	0	0	0	1	2/3
1	0	0	1	1	1/2
1	0	1	0	1	1/2
1	0	1	1	0	1/3
1	1	0	0	0	1
1	1	0	1	0	1
1	1	1	0	0	1
1	1	1	1	0	4/3

$$Y = \overline{A(B + C\dot{D})}$$



$$t_{pLH} \propto \frac{KC}{\beta eq_p} \qquad t_{pHL} \propto \frac{KC}{\beta eq_n}$$

$$\beta eq_p = \frac{\beta_p}{2} = \frac{\mu_p}{2}\frac{W_p}{L_p} \qquad \beta eq_n = \frac{\beta_n}{3} = \frac{\mu_n}{3}\frac{W_n}{L_n}$$

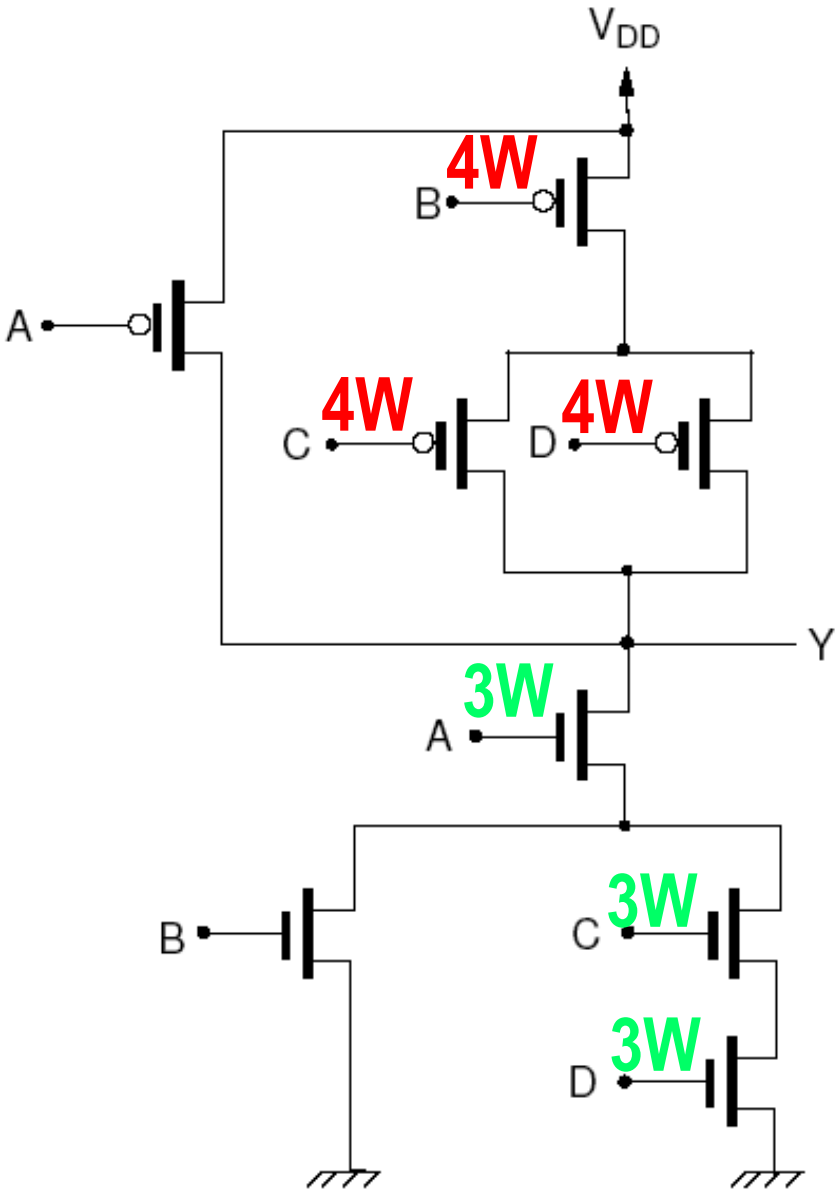
✧ $tp_{HL} = tp_{LH}$ in the worst case

$$\frac{\mu_p}{2}\frac{W_p}{L_p} = \frac{\mu_n}{3}\frac{W_n}{L_n}$$

If $\mu_p = \frac{\mu_n}{2}$ $L_n = L_p = L_{\min}$ → $\frac{W_p}{W_n} = \frac{4}{3}$

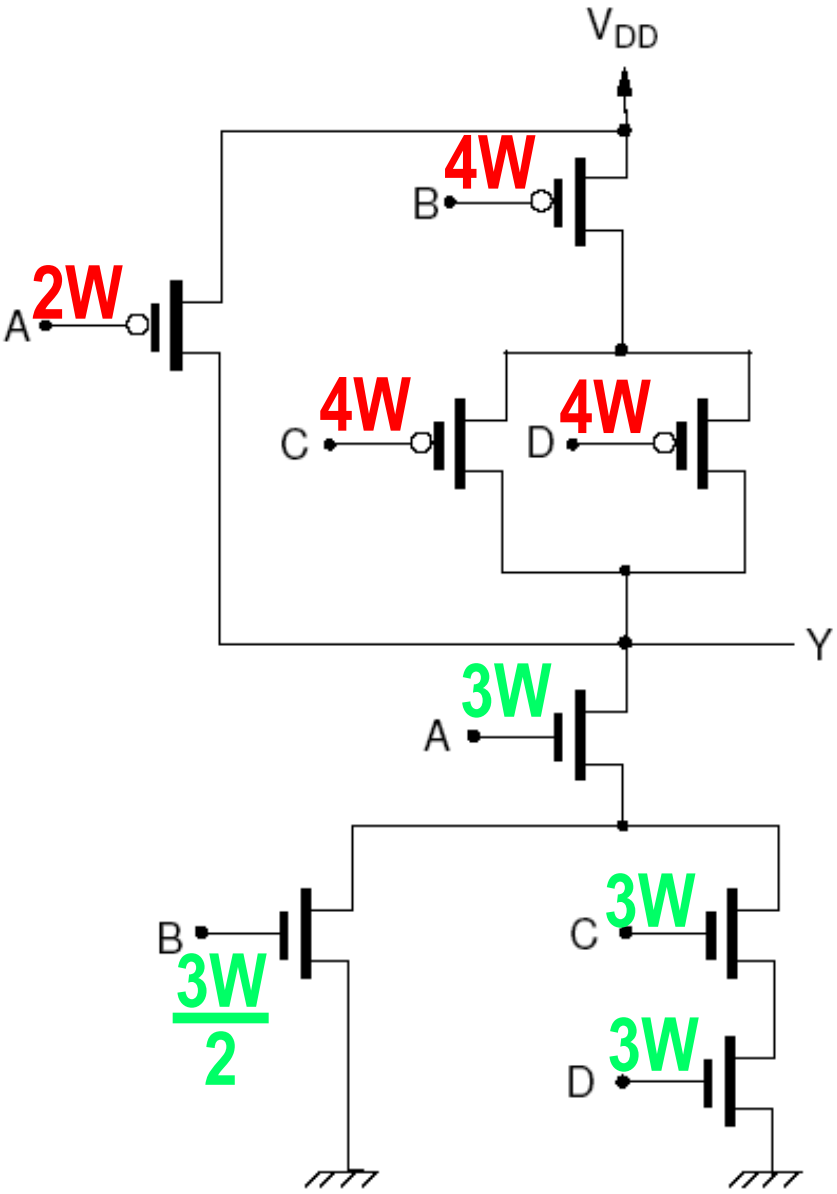
A	B	C	D	Y	ß
0	0	0	0	1	5/3
0	0	0	1	1	3/2
0	0	1	0	1	3/2
0	0	1	1	1	1
0	1	0	0	1	1
0	1	0	1	1	1
0	1	1	0	1	1
0	1	1	1	1	1
1	0	0	0	1	2/3
1	0	0	1	1	1/2
1	0	1	0	1	1/2
1	0	1	1	0	1/3
1	1	0	0	0	1
1	1	0	1	0	1
1	1	1	0	0	1
1	1	1	1	0	4/3

$$Y = \overline{A(B + C\dot{D})}$$

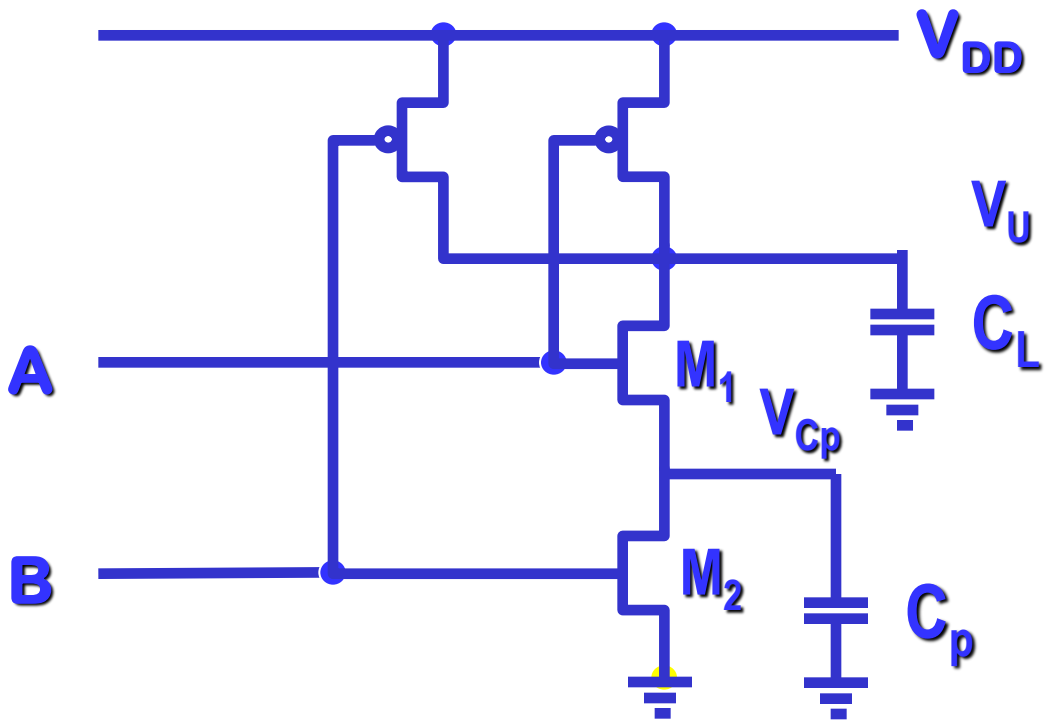


A	B	C	D	Y	ß
0	0	0	0	1	5/3
0	0	0	1	1	3/2
0	0	1	0	1	3/2
0	0	1	1	1	1
0	1	0	0	1	1
0	1	0	1	1	1
0	1	1	0	1	1
0	1	1	1	1	1
1	0	0	0	1	2/3
1	0	0	1	1	1/2
1	0	1	0	1	1/2
1	0	1	1	0	1/3
1	1	0	0	0	1
1	1	0	1	0	1
1	1	1	0	0	1
1	1	1	1	0	4/3

$$Y = \overline{A(B + C\dot{D})}$$



CMOS gate - Body Effect



A = 1 B = ↑

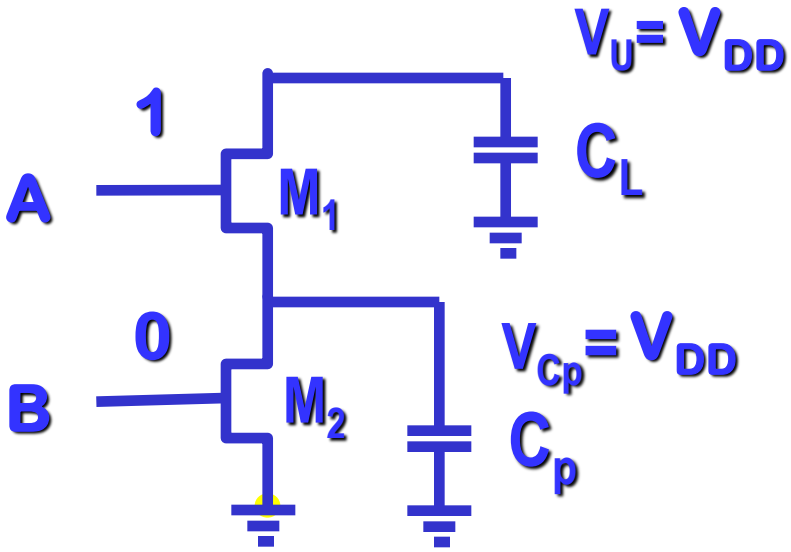
A = ↑ B = 1

A	B	U
0	0	1
0	1	1
1	0	1
1	1	0

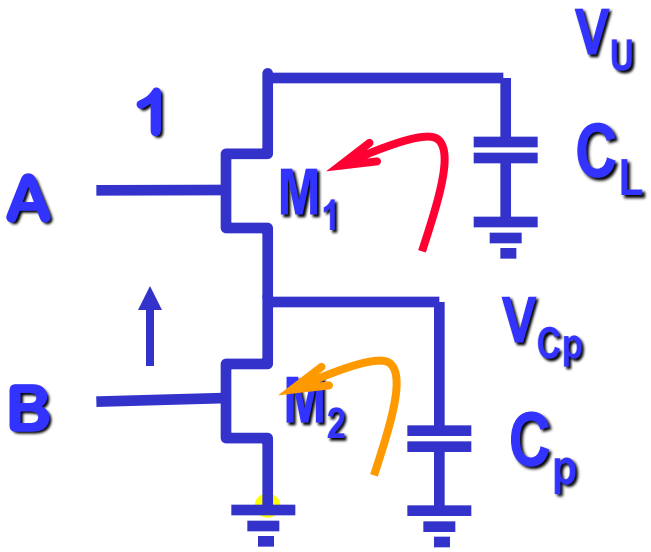
CMOS gate - Body Effect

A = 1 B = 0

t ≤ 0



A = 1 B = ↑

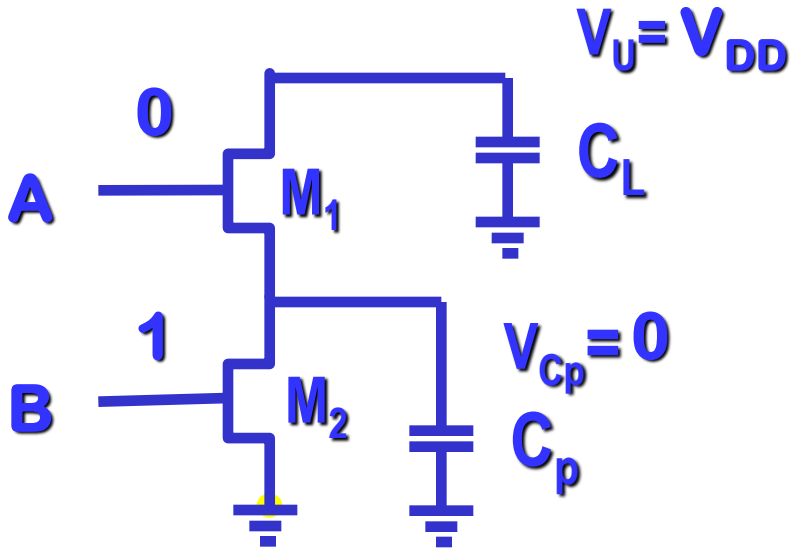


First Cp discharges through M2, M1 starts with Vs=VDD , so low M1 current due to body effect. Longer propagation time w.r.t. to the one estimated with ßeq

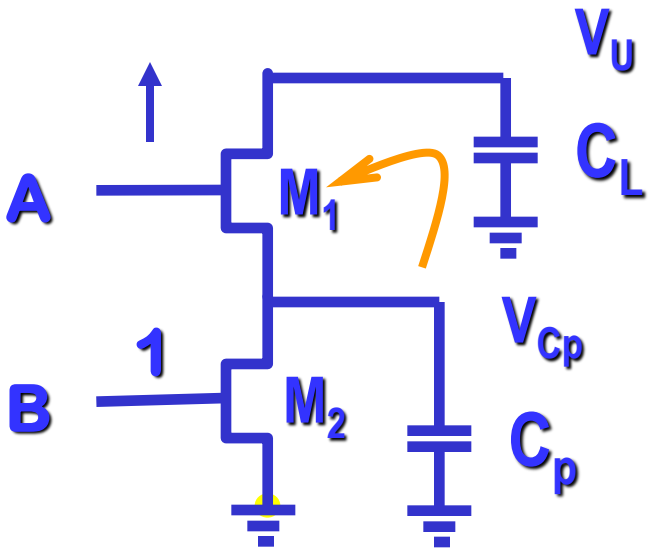
CMOS gate - Body Effect

A = 0 B = 1

t ≤ 0

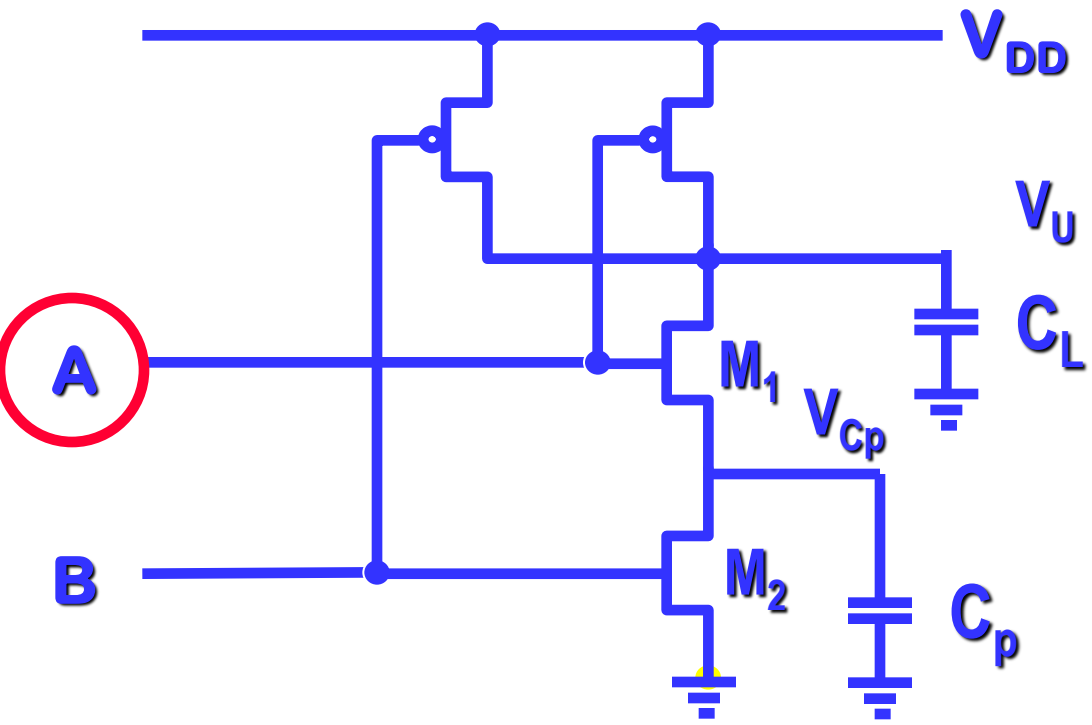


A = ↑ B = 1



Cp is at ground. So after A input swithing, M1 source is at ground through Cp. Reduced body effect and so faster gate swithing.

CMOS gate - Body Effect

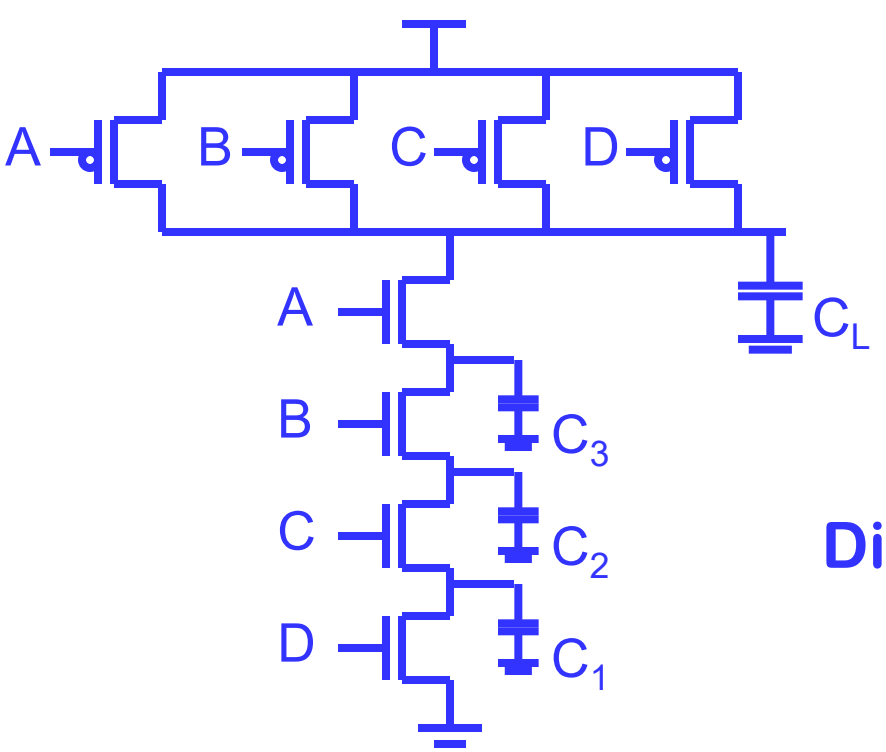


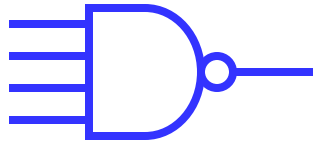
A = 1 B = ↑

A = ↑ B = 1

Faster output swithing are the ones due to swithing of input closer to the output

Dynamic Analysis



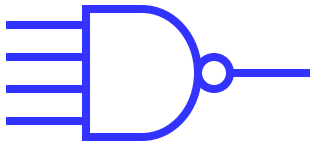


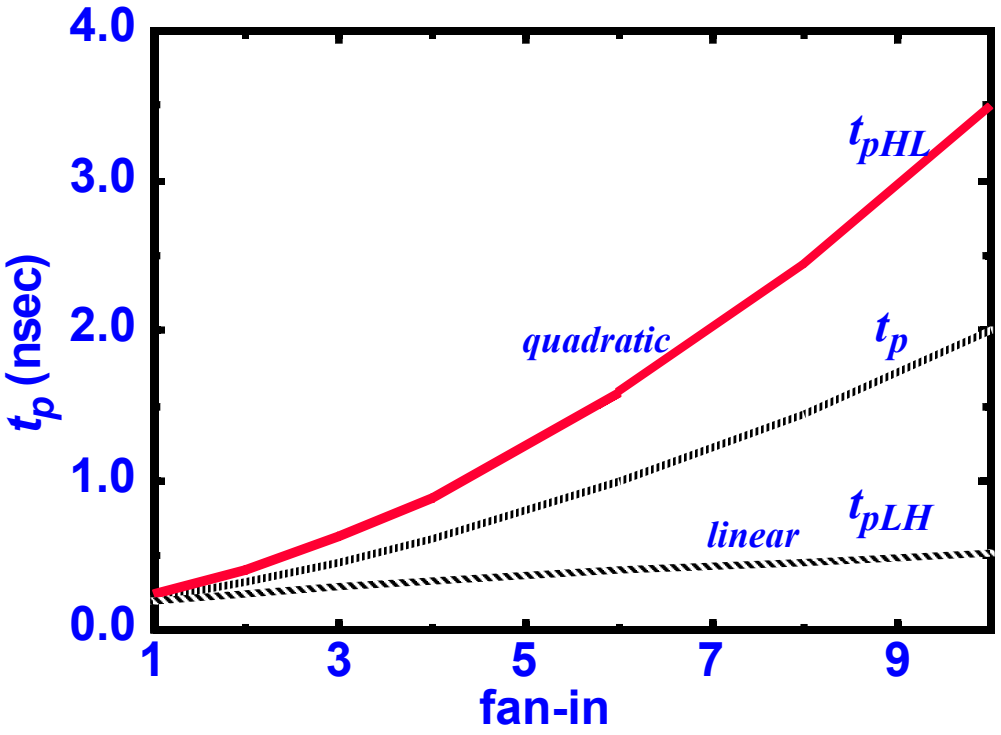
Distributed RC model (Elmore delay)

$$t_{pHL} = 0.69\, R_{eqn}(C_1+2C_2+3C_3+4C_L)$$

Propagation delay deteriorates rapidly as a function of fan-in – quadratically in the worst case

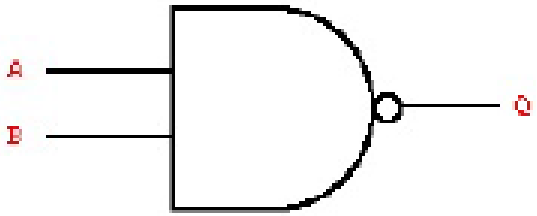
Dynamic Analysis





Gates with fan-in greater than 4 should be avoided

Strength	1
Cell Area	43.680 um^2
Equation	Q = "!(A&B)"
Type	Combinational
Input	A, B
Output	Q



State Table		
A	B	Q
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		5.00	100.00	5.00	100.00
A to Q	fall	0.06	0.58	-0.13	0.95
	rise	0.07	0.68	0.70	1.77
B to Q	fall	0.06	0.58	-0.37	0.60
	rise	0.08	0.69	0.91	1.87

Output Transition [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		5.00	100.00	5.00	100.00
A to Q	fall	0.07	0.83	0.77	1.64
	rise	0.10	1.05	0.75	1.71
B to Q	fall	0.07	0.83	0.76	1.53
	rise	0.11	1.06	0.82	1.72

Capacitance [fF]	
A	2.7240
B	3.0190

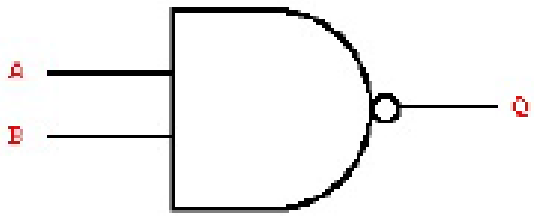
Leakage [pW]
0.28

Dynamic Power Consumption [nW/MHz]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		5.00	100.00	5.00	100.00
A to Q	fall	8.00	8.49	555.73	471.83
	rise	54.46	55.63	732.49	629.72
B to Q	fall	7.37	7.81	616.29	518.84
	rise	64.63	64.73	837.28	703.47

NAND2X1

www.ams.com

Strength	2
Cell Area	43.680 um^2
Equation	Q = "!(A&B)"
Type	Combinational
Input	A, B
Output	Q



State Table		
A	B	Q
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		10.00	200.00	10.00	200.00
A to Q	fall	0.06	0.66	-0.09	1.13
	rise	0.06	0.75	0.61	1.81
B to Q	fall	0.06	0.66	-0.27	0.79
	rise	0.06	0.69	0.75	1.78

Output Transition [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		10.00	200.00	10.00	200.00
A to Q	fall	0.07	0.92	0.75	1.73
	rise	0.08	1.16	0.71	1.80
B to Q	fall	0.07	0.92	0.74	1.61
	rise	0.08	1.07	0.80	1.75

Capacitance [fF]	
A	3.8420
B	4.4020

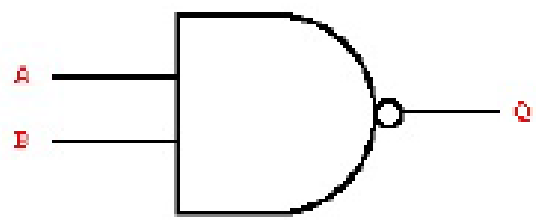
Leakage [pW]
0.31

Dynamic Power Consumption [nW/MHz]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		10.00	200.00	10.00	200.00
A to Q	fall	5.99	6.95	974.79	808.95
	rise	64.40	64.67	1272.31	1068.90
B to Q	fall	4.28	5.14	1162.42	945.27
	rise	77.18	78.81	1504.40	1252.61

NAND2X2

www.ams.com

Strength	6
Cell Area	72.800 um^2
Equation	Q = "!(A&B)"
Type	Combinational
Input	A, B
Output	Q



State Table		
A	B	Q
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		30.00	600.00	30.00	600.00
A to Q	fall	0.05	0.65	-0.04	1.16
	rise	0.05	0.68	0.51	1.68
B to Q	fall	0.05	0.65	-0.28	0.80
	rise	0.06	0.69	0.73	1.77

Output Transition [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		30.00	600.00	30.00	600.00
A to Q	fall	0.06	0.92	0.72	1.70
	rise	0.07	1.06	0.70	1.73
B to Q	fall	0.06	0.92	0.72	1.59
	rise	0.08	1.07	0.78	1.74

Capacitance [fF]	
A	10.1390
B	11.0760

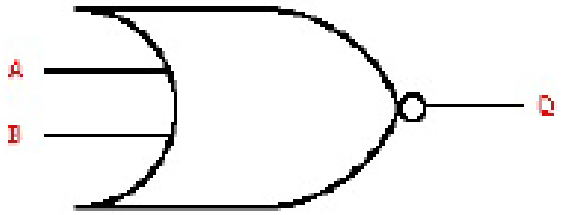
Leakage [pW]
0.35

Dynamic Power Consumption [nW/MHz]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		30.00	600.00	30.00	600.00
A to Q	fall	5.73	8.27	3025.58	2459.24
	rise	155.47	156.48	3806.99	3246.79
B to Q	fall	5.87	8.76	3407.23	2766.61
	rise	196.60	199.91	4408.72	3647.73

NAND2X6

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Strength	1
Cell Area	43.680 um^2
Equation	Q = "!(A | B)"
Type	Combinational
Input	A, B
Output	Q



State Table		
A	B	Q
L	L	H
H	-	L
-	H	L

Propagation Delay [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		5.00	100.00	5.00	100.00
A to Q	fall	0.04	0.41	-0.64	0.48
	rise	0.11	1.17	1.28	2.61
B to Q	fall	0.05	0.42	-0.45	0.56
	rise	0.11	1.18	1.00	2.25

Output Transition [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		5.00	100.00	5.00	100.00
A to Q	fall	0.04	0.55	0.73	1.46
	rise	0.15	1.81	0.62	2.12
B to Q	fall	0.05	0.55	0.94	1.55
	rise	0.15	1.81	0.80	2.12

Capacitance [fF]	
A	2.6610
B	2.9400

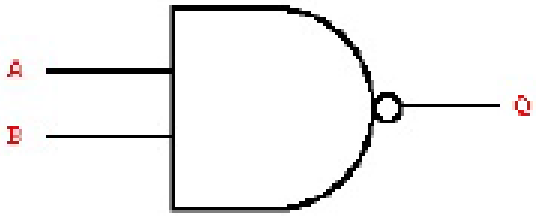
Leakage [pW]
0.26

Dynamic Power Consumption [nW/MHz]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		5.00	100.00	5.00	100.00
A to Q	fall	4.63	5.12	326.27	363.32
	rise	45.80	47.19	661.91	550.88
B to Q	fall	7.48	7.98	453.09	450.46
	rise	55.57	56.53	796.35	658.80

NOR2X1

www.ams.com

Strength	1
Cell Area	43.680 um^2
Equation	Q = "!(A&B)"
Type	Combinational
Input	A, B
Output	Q



State Table		
A	B	Q
L	-	H
H	H	L
-	L	H

Propagation Delay [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		5.00	100.00	5.00	100.00
A to Q	fall	0.06	0.58	-0.13	0.95
	rise	0.07	0.68	0.70	1.77
B to Q	fall	0.06	0.58	-0.37	0.60
	rise	0.08	0.69	0.91	1.87

Output Transition [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		5.00	100.00	5.00	100.00
A to Q	fall	0.07	0.83	0.77	1.64
	rise	0.10	1.05	0.75	1.71
B to Q	fall	0.07	0.83	0.76	1.53
	rise	0.11	1.06	0.82	1.72

Capacitance [fF]	
A	2.7240
B	3.0190

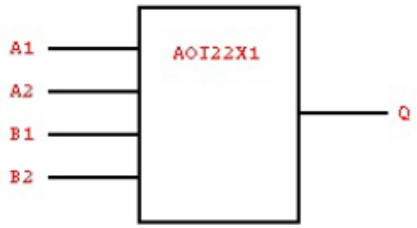
Leakage [pW]
0.28

Dynamic Power Consumption [nW/MHz]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		5.00	100.00	5.00	100.00
A to Q	fall	8.00	8.49	555.73	471.83
	rise	54.46	55.63	732.49	629.72
B to Q	fall	7.37	7.81	616.29	518.84
	rise	64.63	64.73	837.28	703.47

NAND2X1

www.ams.com

Strength	1
Cell Area	72.800 um^2
Equation	Q = "!((A1 &A2) | (B1 &B2))"
Type	Combinational
Input	A1, A2, B1, B2
Output	Q



State Table				
A1	A2	B1	B2	Q
L	-	L	-	H
L	-	-	L	H
H	H	-	-	L
-	L	L	-	H
-	L	-	L	H
-	-	H	H	L

Example

Propagation Delay [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		5.00	100.00	5.00	100.00
A1 to Q	fall	0.06	0.58	-0.27	0.88
	rise	0.11	1.15	1.15	2.47
A2 to Q	fall	0.06	0.58	-0.50	0.54
	rise	0.12	1.16	1.34	2.55
B1 to Q	fall	0.09	0.61	-0.05	0.98
	rise	0.15	1.19	0.85	2.10
B2 to Q	fall	0.09	0.61	-0.31	0.62
	rise	0.16	1.19	1.07	2.20

Output Transition [ns]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		5.00	100.00	5.00	100.00
A1 to Q	fall	0.08	0.81	0.82	1.68
	rise	0.18	1.79	0.73	2.18
A2 to Q	fall	0.08	0.81	0.76	1.53
	rise	0.20	1.81	0.74	2.17
B1 to Q	fall	0.11	0.84	1.08	1.78
	rise	0.18	1.79	0.92	2.19
B2 to Q	fall	0.11	0.84	0.99	1.64
	rise	0.20	1.81	0.96	2.20

Capacitance [fF]	
A1	2.7590
A2	3.0640
B1	2.9460
B2	3.2810

Leakage [pW]
0.40

$$Y = \overline{A_1 \cdot A_2 + B_1 \cdot B_2}$$

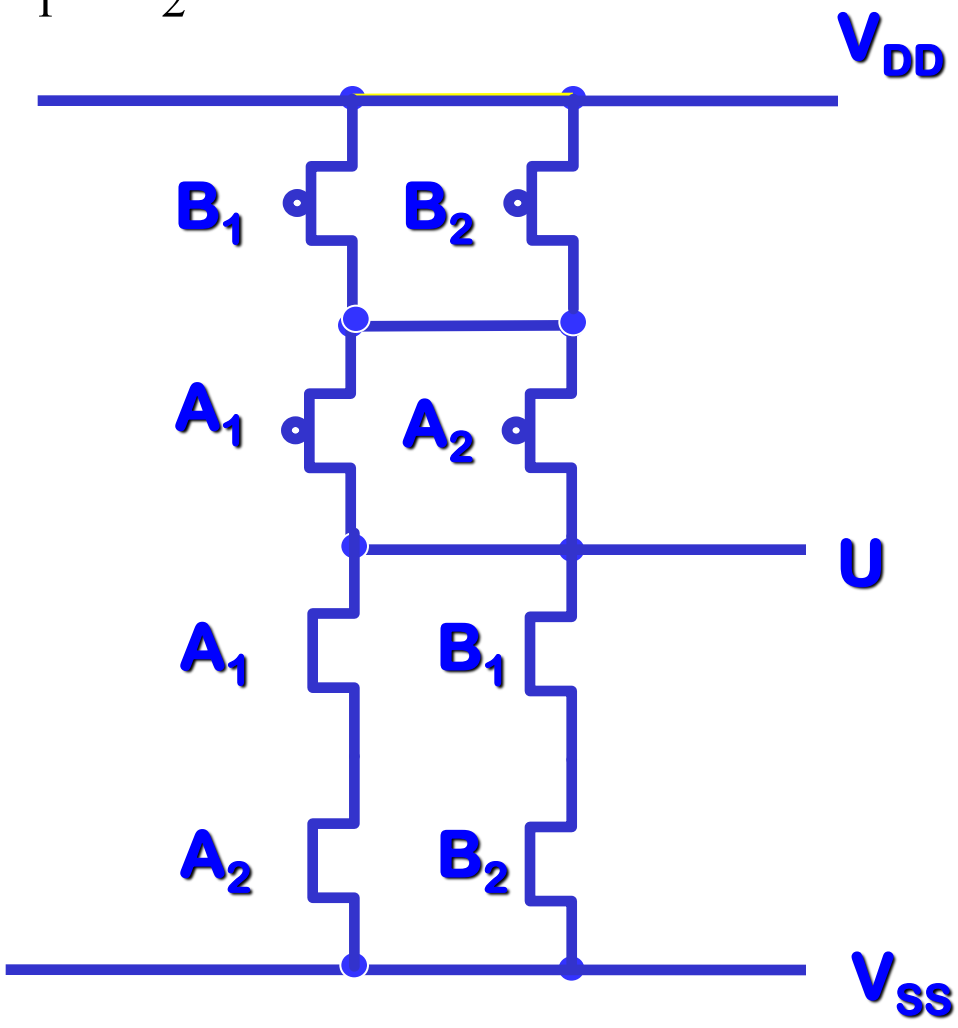
Dynamic Power Consumption [nW/MHz]					
Input Transition [ns]		0.01		4.00	
Load Capacitance [fF]		5.00	100.00	5.00	100.00
A1 to Q	fall	4.83	5.56	391.57	372.83
	rise	52.43	53.92	647.45	530.25
A2 to Q	fall	5.18	5.94	412.34	403.16
	rise	61.25	62.04	717.55	579.91
B1 to Q	fall	34.51	39.00	524.97	460.07
	rise	76.53	77.52	743.28	605.54
B2 to Q	fall	34.58	39.00	569.64	500.92
	rise	84.07	85.01	833.81	670.42

AOI22X1

$$Y = \overline{A_1 \cdot A_2 + B_1 \cdot B_2}$$

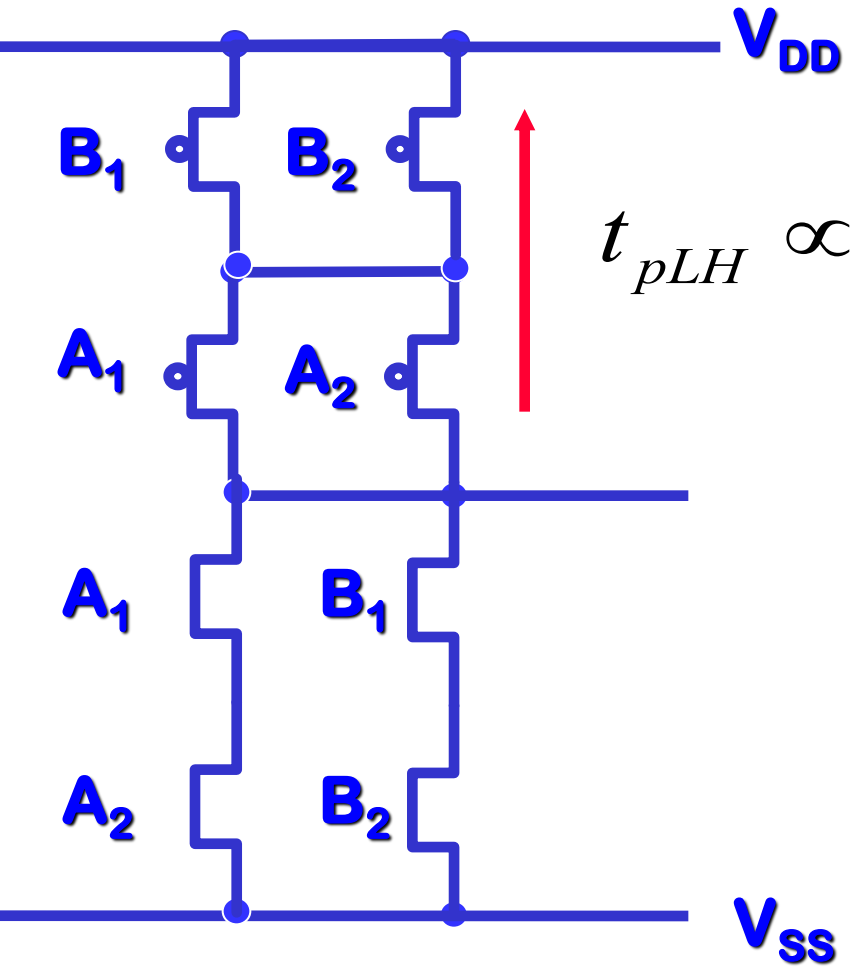
$$Y = \overline{A_1 \cdot A_2 + B_1 \cdot B_2}$$

PDN $\overline{Y} = A_1 \cdot A_2 + B_1 \cdot B_2$



$$Y = \overline{A_1 \cdot A_2 + B_1 \cdot B_2}$$

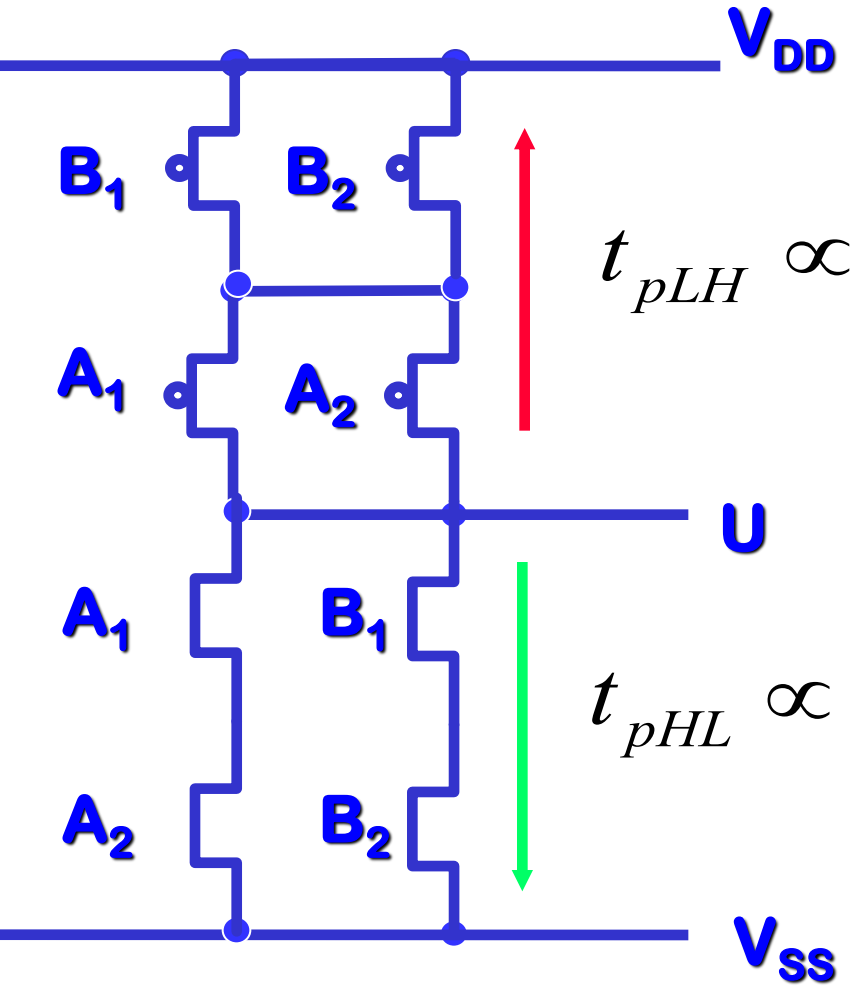
PDN $\overline{Y} = A_1 \cdot A_2 + B_1 \cdot B_2$



$$t_{pLH} \propto \frac{KC}{\beta eq_p} \qquad \beta eq_p = \frac{\beta_p}{2} = \frac{\mu_p}{2}\frac{W_p}{L_p}$$

$$Y = \overline{A_1 \cdot A_2 + B_1 \cdot B_2}$$

PDN $\quad \overline{Y} = A_1 \cdot A_2 + B_1 \cdot B_2$



$$t_{pLH} \propto \frac{KC}{\beta eq_p} \qquad \beta eq_p = \frac{\beta_p}{2} = \frac{\mu_p}{2}\frac{W_p}{L_p}$$

$$t_{pHL} \propto \frac{KC}{\beta eq_n} \qquad \beta eq_n = \frac{\beta_n}{2} = \frac{\mu_n}{2}\frac{W_n}{L_n}$$

$$t_{pLH} \propto \frac{KC}{\beta eq_p} \qquad t_{pHL} \propto \frac{KC}{\beta eq_n}$$

$$\beta eq_p = \frac{\beta_p}{2} = \frac{\mu_p}{2}\frac{W_p}{L_p} \qquad \beta eq_n = \frac{\beta_n}{2} = \frac{\mu_n}{2}\frac{W_n}{L_n}$$

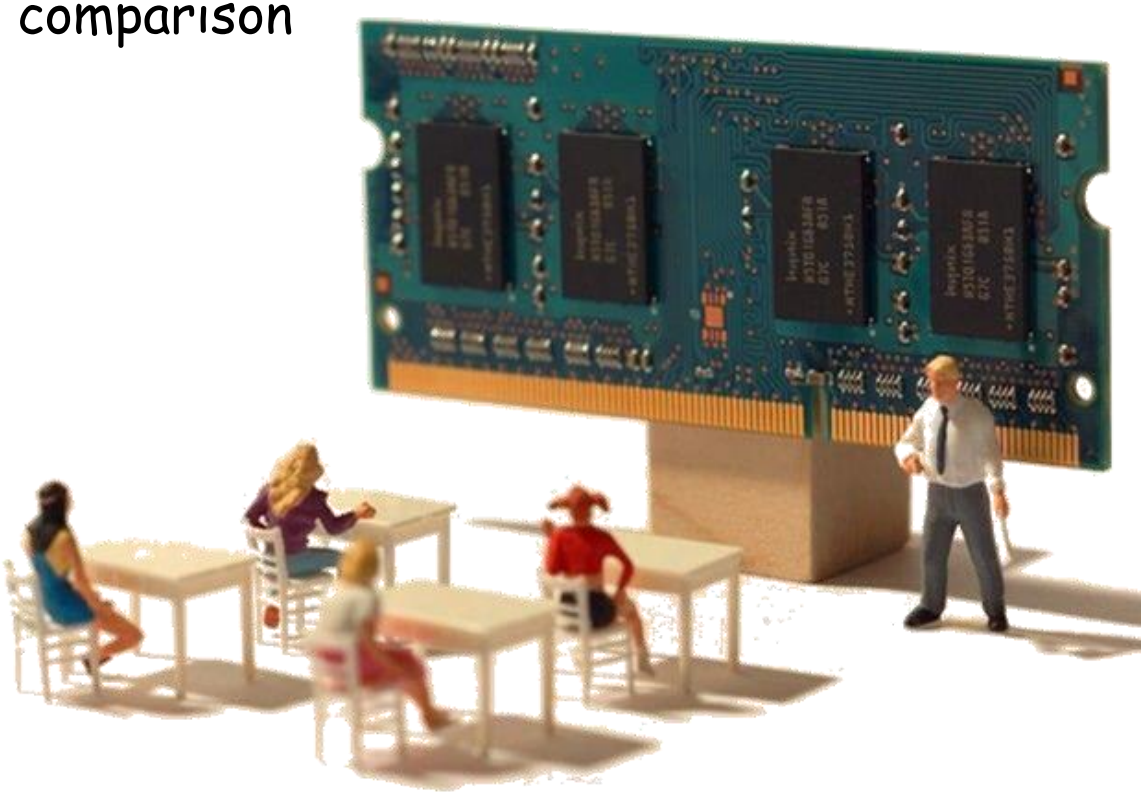
✧ $tp_{HL} = tp_{LH}$ in the worst case

$$\frac{\mu_p}{2}\frac{W_p}{L_p} = \frac{\mu_n}{2}\frac{W_n}{L_n} \qquad \frac{W_p}{W_n} = \frac{\mu_n}{\mu_p}$$

If $\mu_p = \frac{\mu_n}{2}$ $L_n = L_p = L_{\min}$ ⟶ $\frac{W_p}{W_n} = 2$

End Complementary CMOS Logic, Questions ?

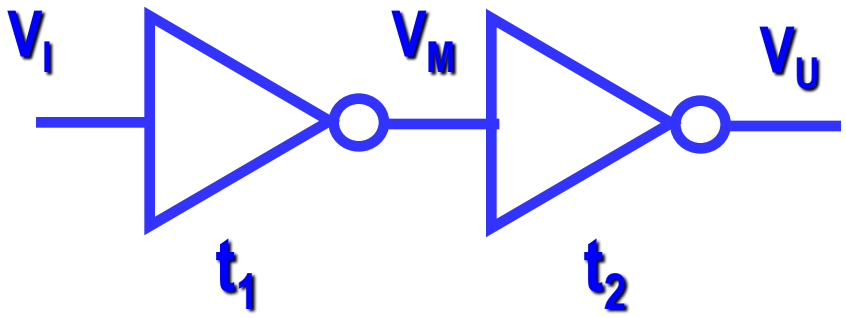
- Synthesis
- Dynamic Analysis
- Examples: NAND, NOR and comparison
- Stick Diagram
- Body Effect
- Standard-cell AMS 0,35 um



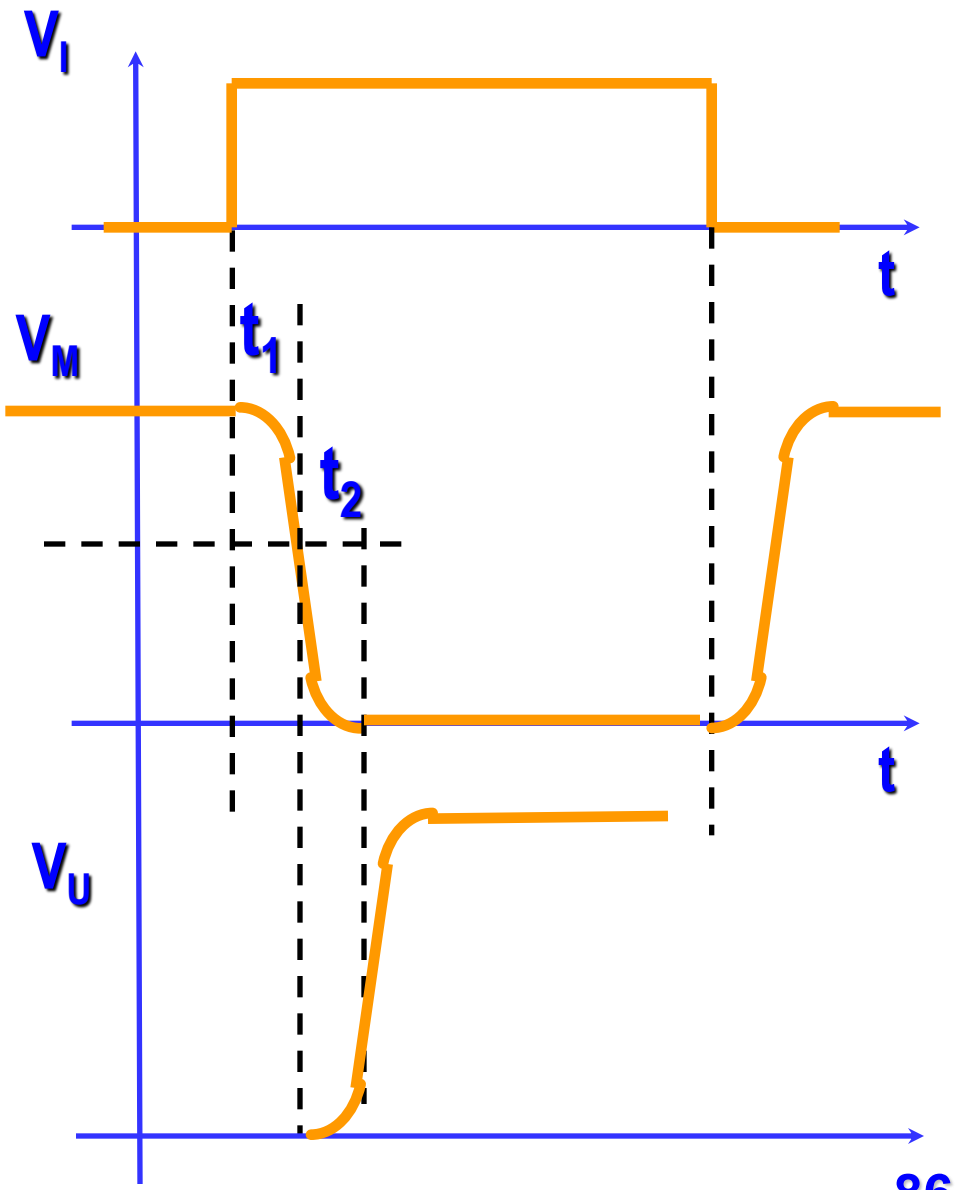
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- Example: Inverter Chain
  - Driving an high value capacitor
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  - Power-Speed Trade-Off
- Example: Pass Transistor Chain
  - Dynamic Analysis
  - Area - Speed Trade-Off

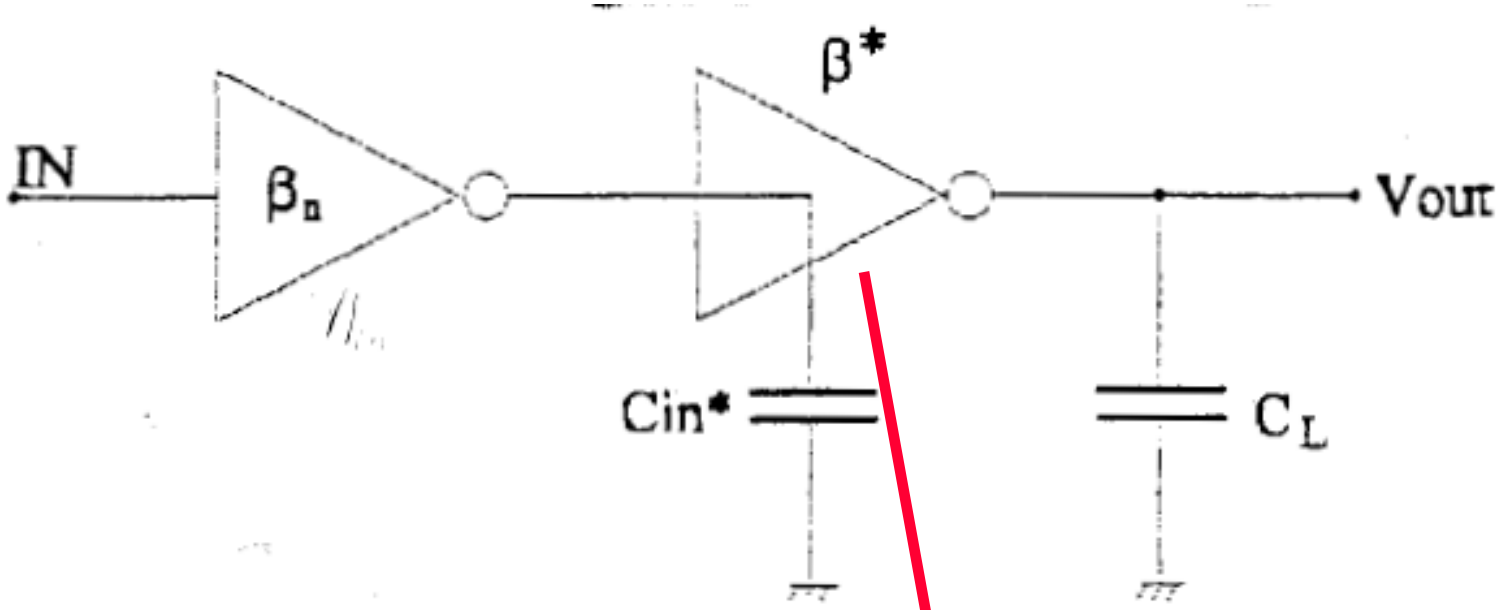
Inverter Chain



$t_{TOT} \sim t_1 + t_2$



Inverter Chain

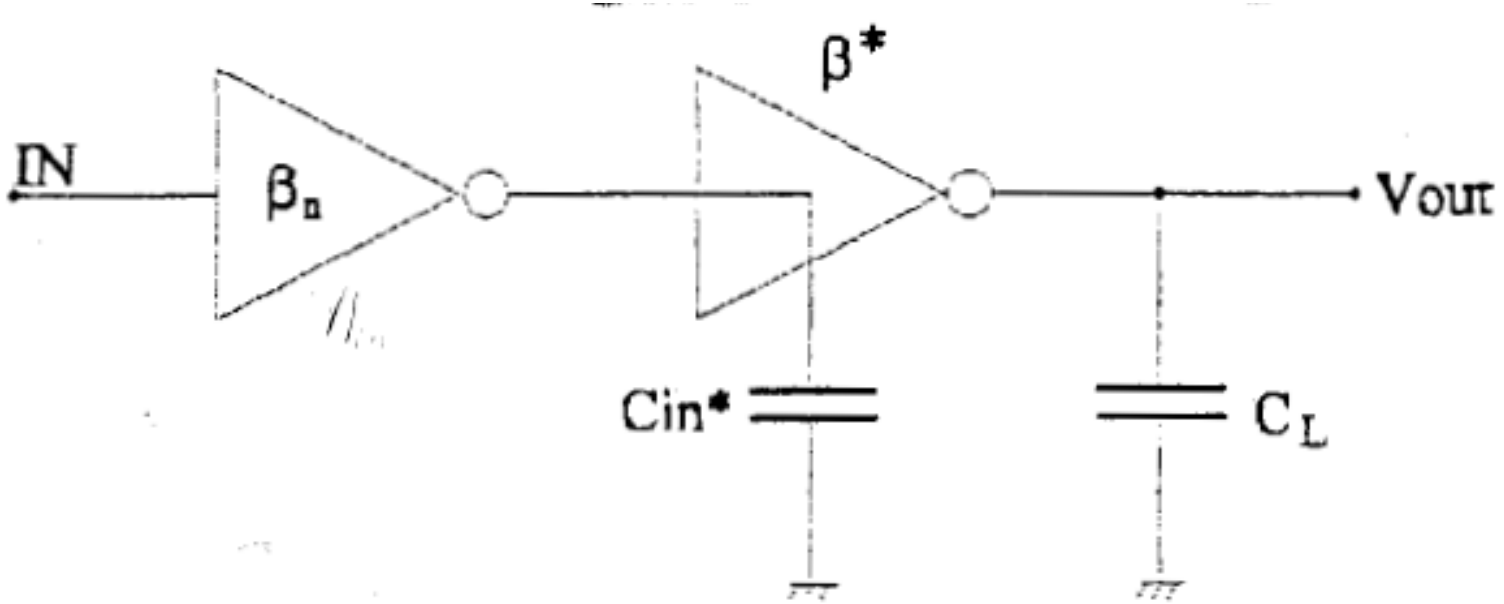


$$t_{phl} = t_{plh} = \frac{KC_{in}}{\beta_n}$$

$$\tau = \frac{KC_L}{\beta^*}$$

Inverter Chain

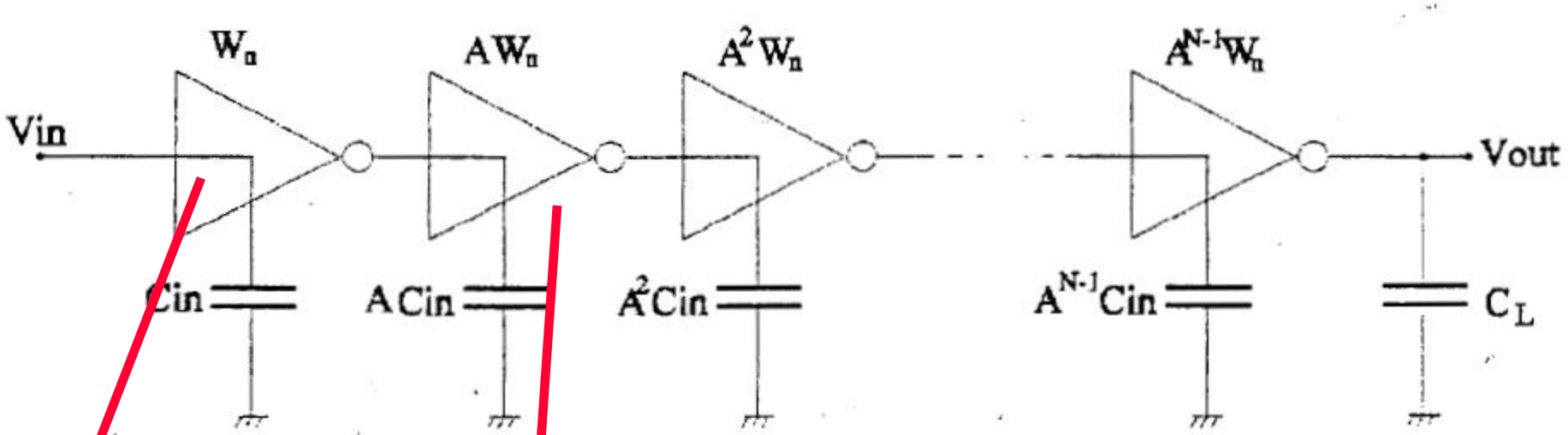
Inverter Chain



$$\tau = \frac{KC_L}{\beta^*} \qquad \frac{KC_{in}}{\beta_n} = \frac{KC_L}{\beta^*} \Rightarrow \beta^* = \frac{C_L \beta_n}{C_{in}} \approx 1000\beta_n$$

$$W^* = \frac{W_n C_L}{C_{in}} \approx 1000 W_n$$

Inverter Chain



$$\tau_1 = K\frac{ACin}{\beta_n} = A\tau_m \qquad \tau_2 = K\frac{A^2Cin}{A\beta_n} = A\tau_m = \tau_1$$

$$\tau_i = K\frac{A^iCin}{A^{i-1}\beta_n} = A\tau_m = \tau_1 \qquad \tau_{tot} = NA\tau_m$$

$$C_L = A^N C_{in}$$

$$\tau_m = \frac{KC_{in}}{\beta_n}$$

$$\tau_{tot} = \frac{A}{\ln A}\tau_m \ln\frac{C_L}{Cin}$$

Inverter Chain

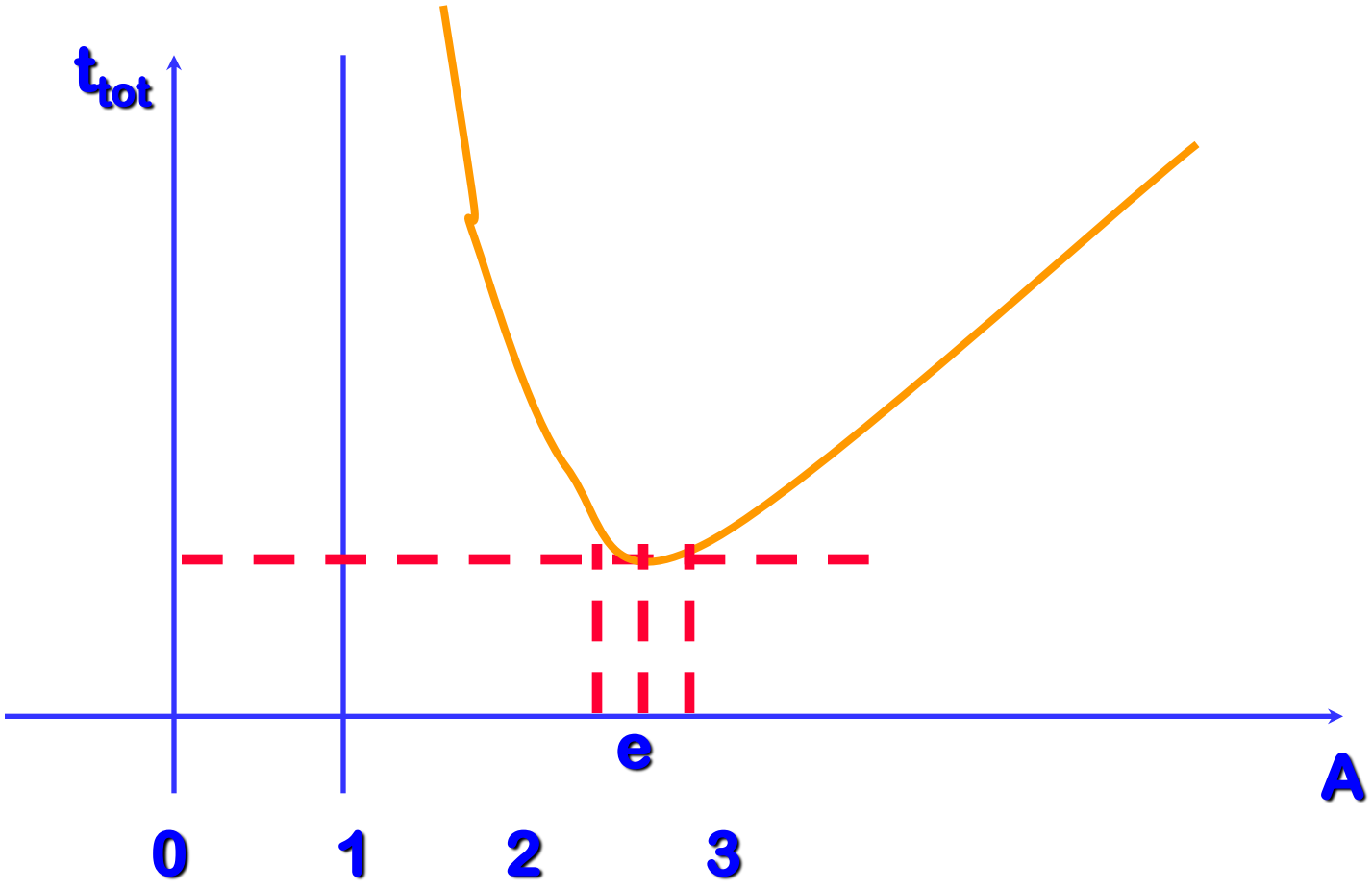
$$\tau_{tot} = \frac{A}{\ln A}\tau_m \ln\frac{C_L}{Cin} \qquad \frac{\partial \tau_{tot}}{\partial A} = \frac{1-\ln A}{\ln^2 A}\tau_m \ln\frac{C_L}{Cin}$$

$1 - \ln A = 0$ **and so** $A = e\,(= 2.71\ldots)$

$$C_{in} = 10\,fF, \qquad C_L = 10\,pF, \qquad N = \frac{\ln(1000)}{\ln(3)} = 6{,}29 \qquad \rightarrow 7$$

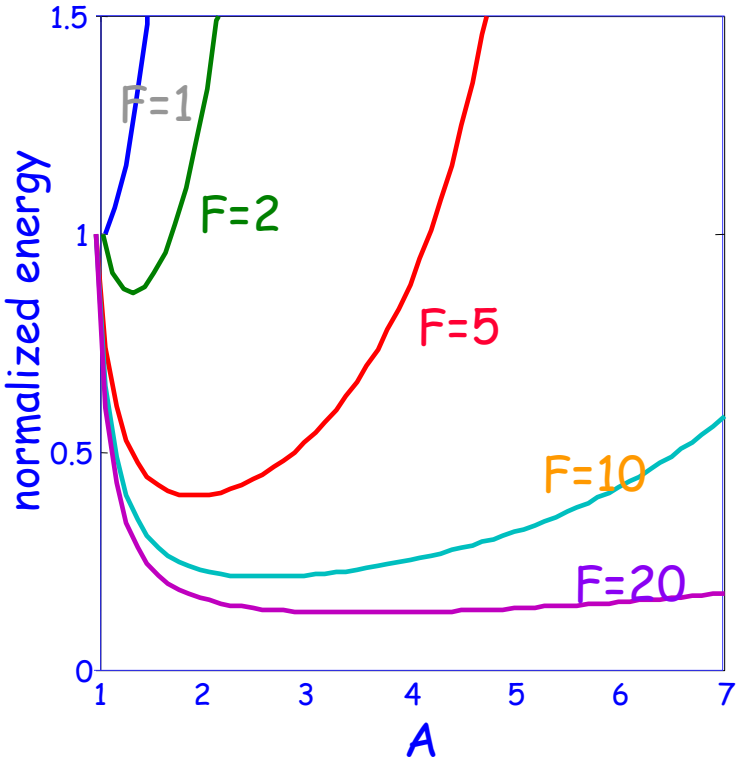
$$\tau_{tot} = 21\tau_m \qquad w.r.t. \qquad \tau_{tot} = 1000\tau_m$$

Inverter Chain



Dynamic Power as a Function of Device Size

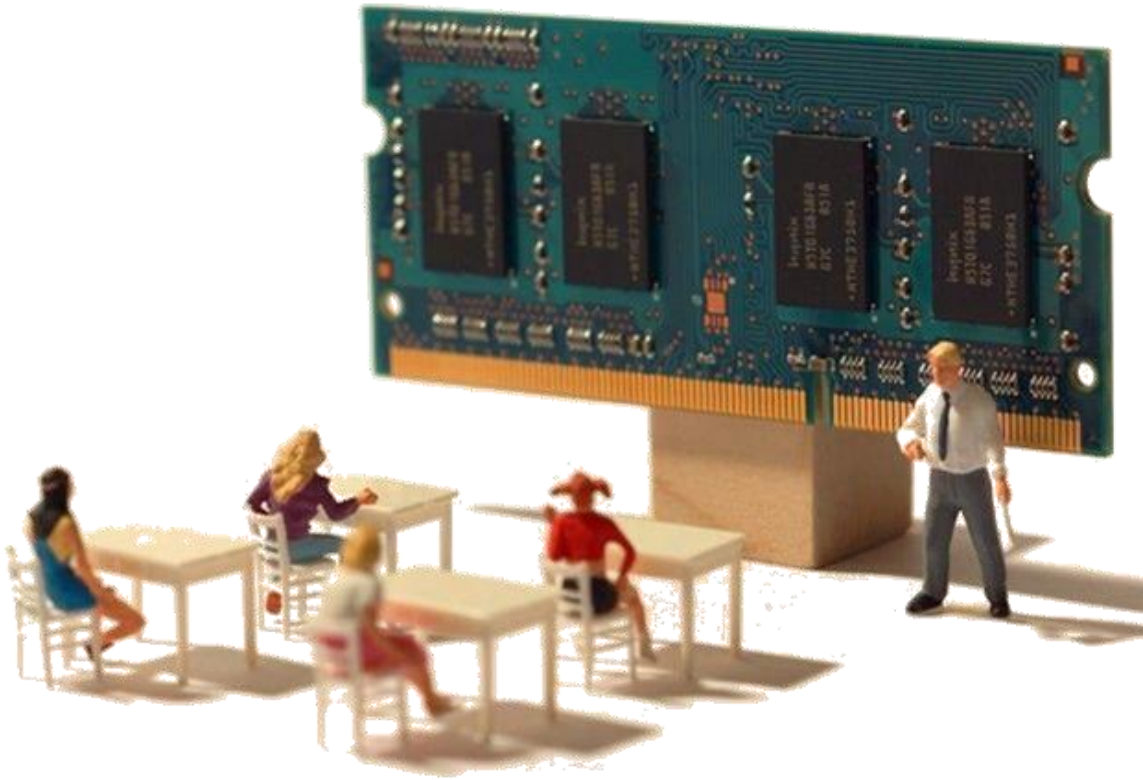
- Device sizing affects dynamic energy consumption
  - gain is largest for networks with large overall effective fan-outs ($F = C_L/C_{in}$)
- The optimal gate sizing factor (A) for dynamic energy is smaller than the one for performance, especially for large F's
  - e.g., for F=20,
    A_{opt}(energy) = 3.53 while
    A_{opt}(performance) = 4.47
- If energy is a concern avoid oversizing beyond the optimal



From Nikolic, UCB

End Inverter Chain Example, Questions ?

- Driving an high value capacitor
- Dynamic Analysis
- Power-Speed Trade-Off

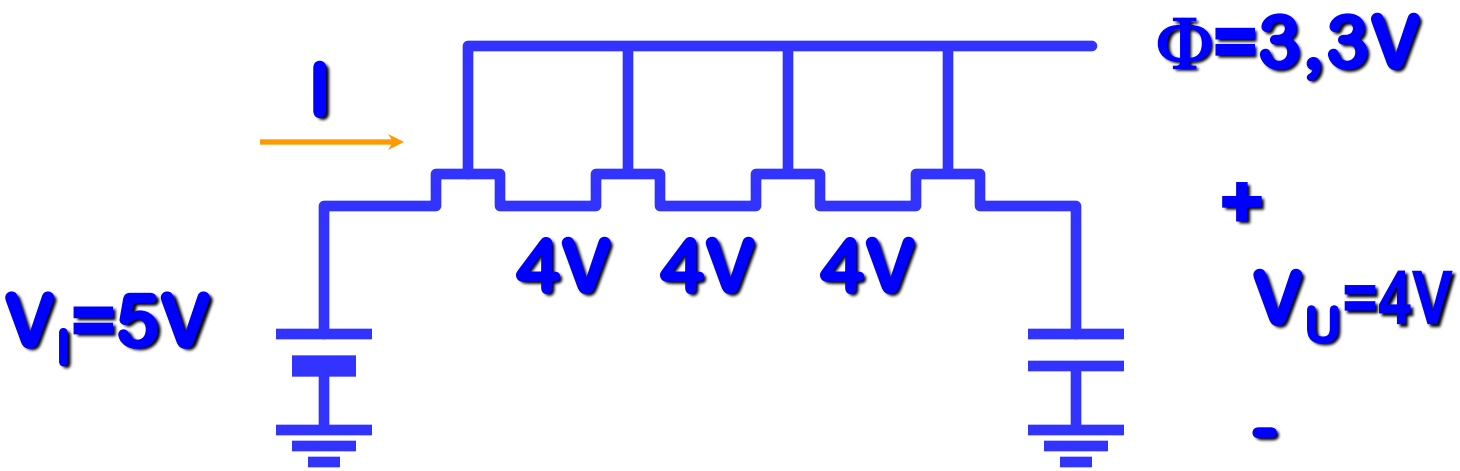


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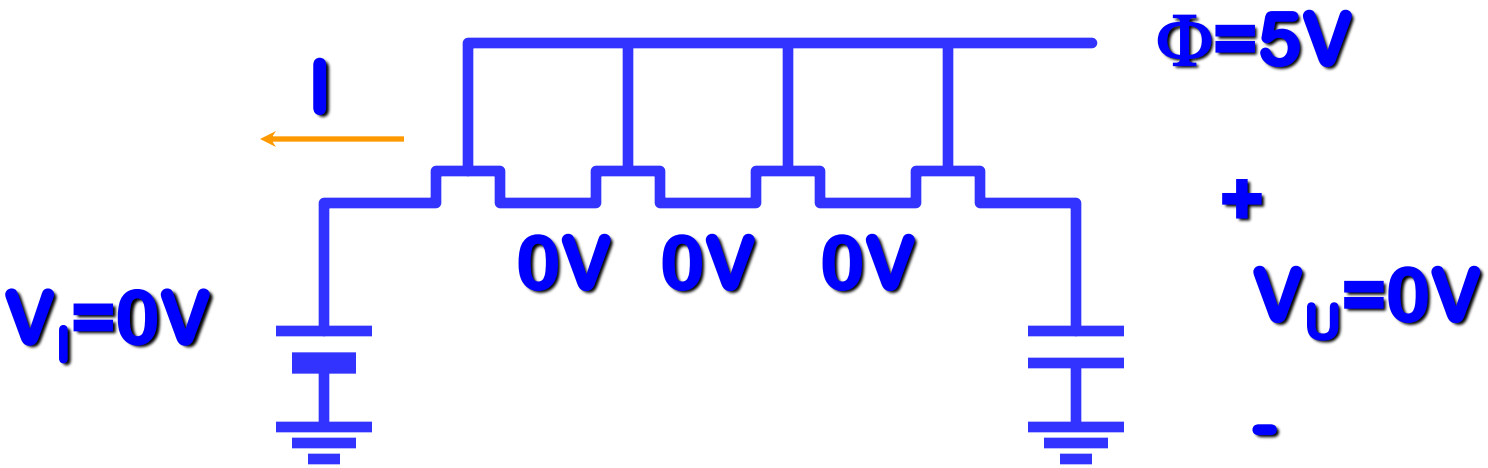
N pass transistor chain

- Threshold Drop at V_H

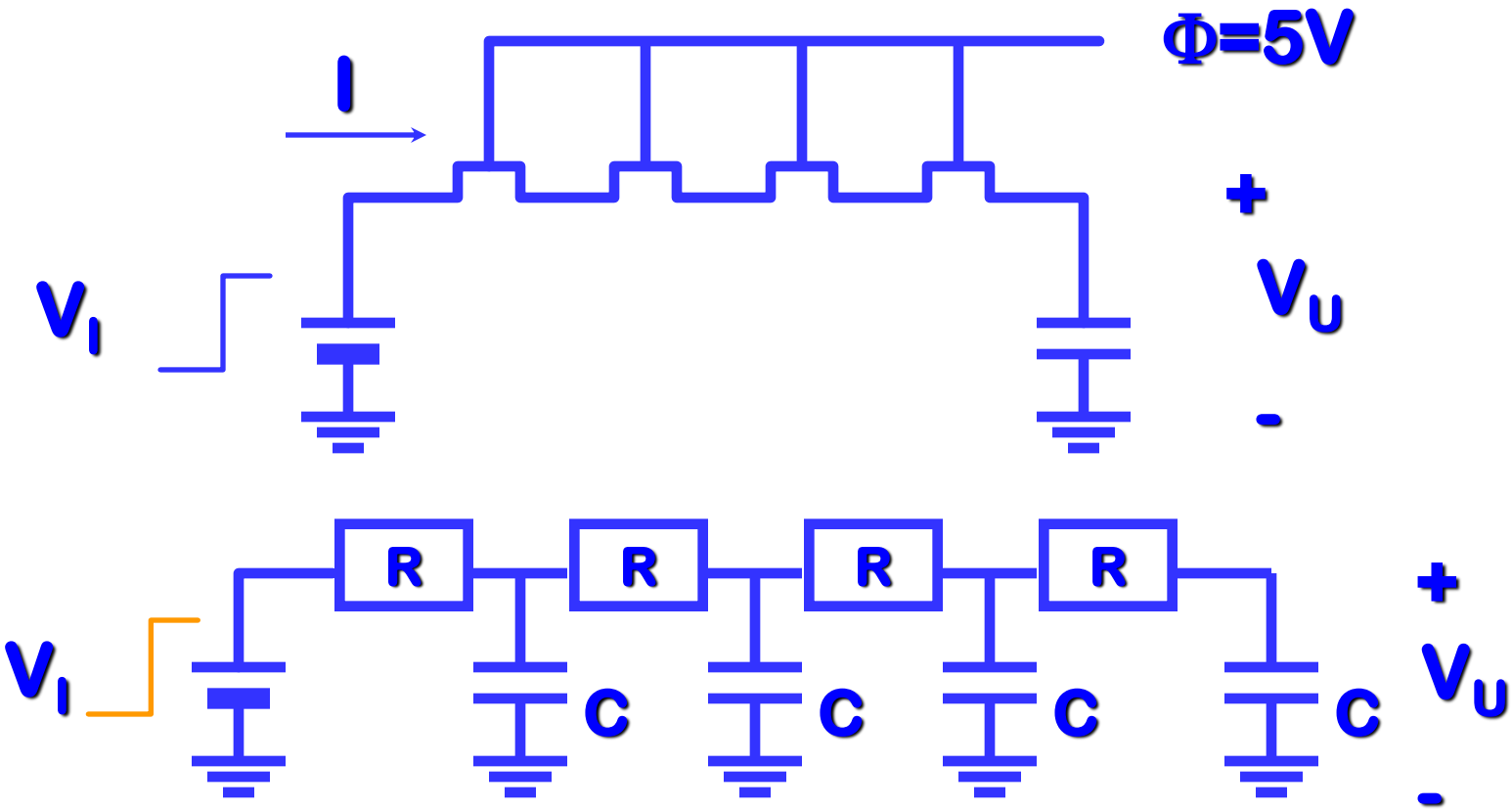


N pass transistor chain

- No Threshold Drop at V_L



N pass transistor chain
Dynamic Analysys

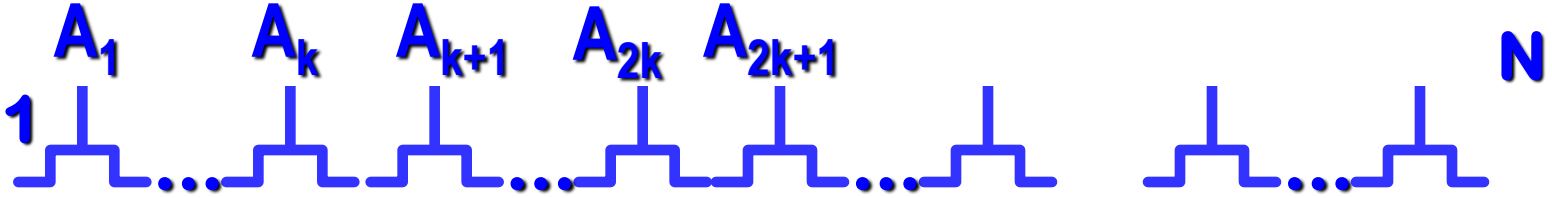


Delay

- T_p = delay of one R-C cell
- Overall delay grows with N^2
- $T_{TOT} = T_p N^2$
- Overall delay becomes very high just after few pass transistors

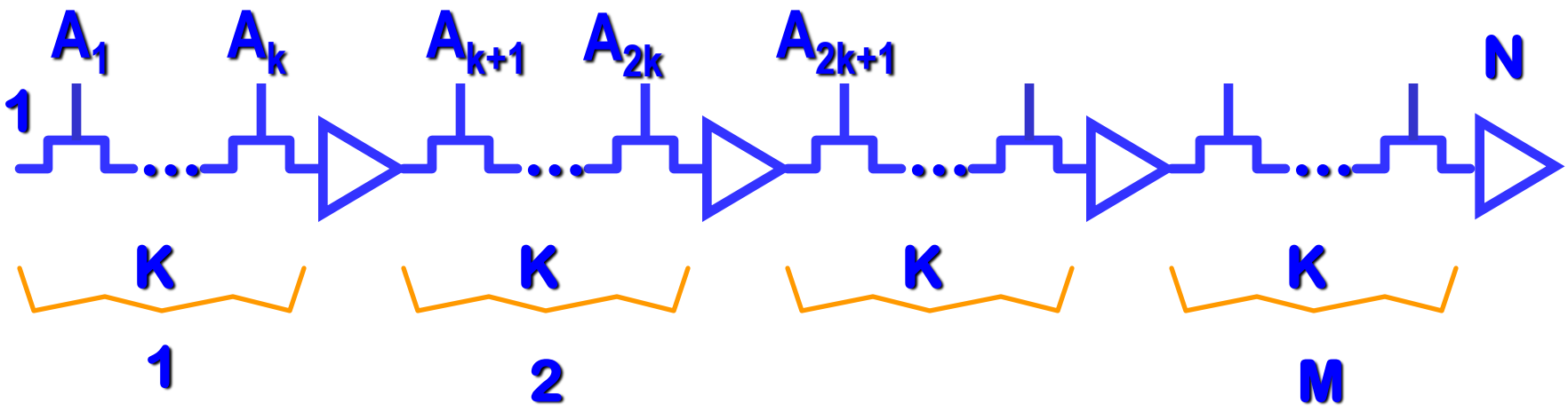
Delay Optimization

- Note:
  - Delay in a buffer chain grows linearly
- Solution
  - Introduce buffers in the pass transistor chain
- Let's consider a N pass transistor chain with « N » elements



Circuit Solution

- T_P = pass transistor delay
- T_B = buffer delay
- M groups made by K pass transistors



Analysis

$$N = K \bullet M; \qquad M = \frac{N}{K}$$

$$T_{tot} = MT_B + M\left(K^2 T_P\right) = \frac{N}{K} T_B + \frac{N}{K}\left(K^2 T_P\right) = \frac{N}{K} T_B + NKT_P$$

$$\frac{dT_{tot}}{dK} = -\frac{N}{K^2} T_B + NT_P = 0$$

$$K = \sqrt{\frac{T_B}{T_p}}$$

Example

✧Example: 16 Bit Manchester Carry Adder

$$N = 16 \qquad T_B = 4ns \qquad T_P = 0{,}2ns$$

$$K = \sqrt{\frac{T_B}{T_p}} = \qquad 4{,}47 \Rightarrow 4 \Rightarrow M = 16 : 4 = 4$$

$$T_{tot} = 4T_B + 4(16T_P) = 28{,}8ns$$

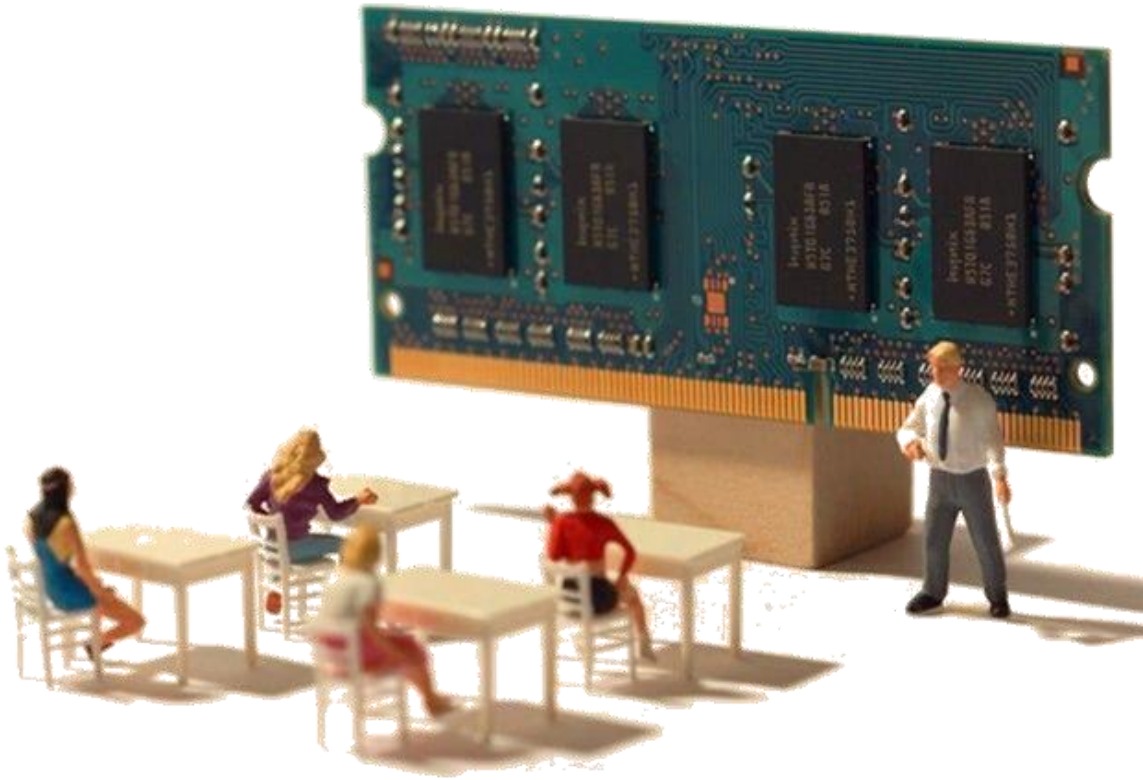
$$w.r.t. \qquad N^2 T_P = 51{,}2ns$$

Pass Gate Chain

- No Threshold drop
- Pass gate delay is lower than pass transistor one
- Delay optimization works as just shown for the pass transistor chain

End Pass Transistor Chain Example, Questions ?

- Dynamic Analysis
- Area - Speed Trade-Off



End, Questions ?

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